

Cross-Country Comovement of Momentum Returns

*Andy Naranjo and Burt Porter**

Latest Version: July 2004

*University of Florida
Warrington College of Business
Department of Finance
P.O. Box 117168
Gainesville, FL 32611-7168
Phone: (352) 392-0153
Fax: (352) 392-0301

We thank Mark Flannery, Kent Hargis, Jason Karceski, Andrew Karolyi, M. Nimalendran, and seminar participants at the University of Miami, University of New Orleans, and CIBER at the University of Florida for helpful comments and suggestions.

Cross-Country Comovement of Momentum Returns

Abstract

This paper examines the sources of cross-country comovement of momentum returns over the 1975-2002 period. Using data on more than 16,000 individual firms across 100 industries from 38 countries, we document the profitability of momentum trading strategies using individual firm returns, industry returns, and industry-adjusted returns. We show that country-neutral momentum returns are significantly correlated across countries and are time-varying. We find that although both across-industry and within-industry momentum is profitable in a large number of countries, comovement among industries cannot explain the comovement of country-neutral momentum returns. However, we find that standard risk factor models do explain a significant portion of the cross-country comovement of momentum returns, even though they do not explain average momentum returns.

I. Introduction

Over the past decade, numerous researchers have documented the profitability of momentum trading strategies in both the U.S. and many other countries around the world (e.g., Jegadeesh and Titman (1993, 2001) Rouwenhorst (1998, 1999), and Chui, Titman and Wei (2000)). While there is strong evidence that momentum strategies earn positive profits, there is little consensus as to the source of these profits. A closely related puzzle is the comovement of country-neutral momentum returns.¹ Rouwenhorst (1998) finds that European and U.S. momentum returns have a correlation of 0.43 during the 1980-1995 period and that by conditioning the returns of a European momentum strategy on the returns to a U.S.-only strategy, the average return of the European momentum portfolio is reduced from 0.93 to 0.65 percent per month, implying a common component to the two series.² Rouwenhorst (1998), however, does not examine the source or dynamics of the comovement. A better understanding of the comovement of country-neutral momentum returns is important because it can potentially provide some evidence as to why momentum strategies earn positive average returns and because the comovement impacts the volatility of momentum portfolios. In this paper, we examine the sources of the cross-country comovement in momentum returns and find that risk factors play an important role in explaining the comovement, even though they are unable to explain the level of momentum returns.

In recent work on the general comovement of security returns, Barberis, Shleifer, and Wurgler (2004) argue that there are three explanations for the observed comovement among different securities: fundamentals-induced, category-induced, and habitat-induced comovement. The fundamental approach explains the comovement of securities by positive correlations in the rational determinants of their values. Category-induced comovement, on the other hand, occurs when investors classify different securities into the same asset class and shift resources in and out of this class in correlated ways.

¹ Country-neutral momentum is defined as zero-cost investment strategy long recent winners and short recent losers where winners and losers are defined relative to all stocks *within* a specific country.

² In contrast to Rouwenhorst (1998), Griffin, Ji, and Martin (2003) report low intraregional and interregional correlations of momentum profits, although these correlations are not the focus of their paper. However, we find that the averaging of country-neutral momentum profits within and across regions, across developed and emerging markets, and over time significantly attenuates country-neutral momentum return correlations resulting in low correlations similar to those reported by Griffin, Ji, and Martin.

Similarly, habitat-induced comovement arises when a group of investors restricts its trading to a given set of securities and moves in and out of that set in tandem. These three explanations suggest that comovement of securities essentially arises from fundamental factors, behavioral/mechanically-induced factors, or a combination of both.

The comovement explanations proposed by Barberis, Shleifer, and Wurgler (2004) suggest that the payoffs of country-neutral momentum portfolios should be correlated if momentum strategies are exposed to some common, yet unidentified priced risk factor and markets are integrated. Alternatively, the comovement in country-neutral momentum returns could arise from the habit or category-induced comovement caused by institutional traders pursuing international momentum trading strategies across countries or within industries across countries. A pervasive behavioral bias might also induce comovement in momentum returns, especially if information flows to which investors over/under react are correlated across markets.

To investigate the sources of cross-country comovement of momentum returns, we assemble a large data set that consists of more than 16,000 individual firms across 100 industries from 38 developed and emerging markets over the 1975 – 2002 period. Using momentum portfolio sorts, we examine the profitability and cross-country comovement of unadjusted, industry-adjusted, and risk-adjusted momentum returns both across countries and over time.

We show that country-neutral momentum strategies are profitable for both developed and emerging markets, yielding an average of 62 and 92 basis points per month respectively over the 1975 – 2002 sample period. This result is consistent with the international evidence presented by Rouwenhorst (1998, 1999) and Griffin, Ji, and Martin (2003). We also find both an industry momentum effect and a firm momentum effect independent of its industry for individual countries and on average across developed countries using country-neutral momentum strategies. In contrast, for the emerging market countries, we do not find significant industry momentum effects, though we do find significant firm effects independent of its industry on average. Using both domestic and global versions of market models and Fama-French three factor models, we confirm that simple risk factor models generally do a poor job of explaining the level of momentum returns in both developed and emerging market countries.

Interestingly, we find that country-neutral momentum returns are significantly related across countries. This relationship is positive and time varying, with a small average correlation in the early 1980s and an increasing average correlation in the 1990s through early 2000s. Conditioning on industry versus firm momentum effects, we find that comovement among industries cannot explain the comovement of country-neutral momentum returns. However, we do provide evidence supporting a simple explanation for the comovement of country-neutral momentum returns. We find that the cross-country comovement of momentum returns is explained by positive correlations in the rational determinants of their values, even though these very same rational determinants are unable to explain the *level* of momentum returns. That is, the correlation of known risk factor payoffs across countries induces correlated time varying factor loadings in country-neutral momentum strategies that in turn induce cross country correlation in momentum portfolio payoffs. Even though standard factor models are unsuccessful in explaining average momentum returns, they are successful in explaining a large part of the cross-country comovement of momentum returns. Taken together, our results suggest that risk factors play an important role in explaining the comovement of country-neutral momentum returns.

The rest of the paper proceeds as follows. Section II provides a brief review of the momentum literature. Section III describes our methodology and data. Section IV documents the cross-country evidence on the profitability of country-neutral momentum returns, while Section V examines cross-country comovement of momentum returns. Section VI provides evidence on risk factor and momentum return comovement across countries. Section VII concludes.

II. Momentum Returns and Comovement: The Literature

Many researchers have documented the profitability of momentum trading strategies in both developed and emerging markets. For example, using U.S. data, Jegadeesh and Titman (1993) find that a strategy that is long recent winners and short recent losers earns statistically significant average returns of 0.95% per month over the 1965 – 1989 period. In later work, Jegadeesh and Titman (2001) find similar results using U.S. data from 1990-1998, suggesting the earlier result was not a manifestation of

data mining. Looking for momentum return evidence across other countries, Rouwenhorst (1998) finds significant momentum profits for 11 of 12 European countries, Chui, Titman, and Wei (2000) find mixed results for 8 Asian countries, and Rowenhorst (1999) finds mixed results for 20 emerging markets.³ Rouwenhorst (1998) also finds that country-neutral momentum returns are highly correlated across markets.

Much research attention has also been given to explaining why momentum trading strategies are so widely profitable. Using a simple framework, Jegadeesh (1990), Lo and MacKinlay (1990), and Jegadeesh and Titman (1993) show that the source of momentum profits can be decomposed into three different components: the cross-sectional dispersion in expected returns, momentum in systematic factors, and momentum in the idiosyncratic component of security returns (i.e., “over-reaction/under-reaction component”). With this framework as a backdrop, two lines of research have emerged in the investigation of momentum profits: one based on rational models and the other based on behavioral models. These two lines of research are also important for understanding the potential sources of comovement in momentum returns.

Within the rational paradigm, the profitability of moment strategies arises from compensation for risk. Conrad and Kaul (1998) argue that momentum profits reflect cross-sectional variation in expected returns rather than predictable time-series variation in stock returns. They argue that the momentum strategy’s average profits reflect the result of buying (on average) high mean return stocks and selling (on average) low mean return stocks. If the cross-sectional differences in the mean returns reflect expected returns, then the momentum profits can be attributed to cross-sectional differences in risk characteristics (i.e., common identified or yet unidentified priced risk factors).⁴ If

³ While momentum strategies have been found to be profitable for medium-term holding periods (3 to 12 months), another strand of the literature finds that contrarian strategies have been somewhat profitable for short-term (less than 1 month) and long-term (3 to 5 years) intervals (e.g., DeBondt and Thaler (1985) and Jegadeesh (1990)). The long-term contrarian strategy profits are largely limited to pre-1947 data, while the short-term contrarian profits are often generated by microstructure biases (e.g., bid-ask bounce and inventory effects). See, for instance, Fama and French (1988), Kim, Nelson and Startz (1991), Conrad, Gultekin, and Kaul (1997), Conrad and Kaul (1998), and Asness, Porter and Stevens (2001).

⁴ Many researchers have examined various factors as potential explanations for momentum profits. These include market beta, firm size, book-to-market, cash-flow-to-price, business cycles, and macroeconomic risks among others (see, for instance, Jegadeesh and Titman (1993), Fama and French (1996), Carhart (1997), Daniel, Grinblatt, Titman, and Wermers (1997), Grinblatt and Moskowitz (1999), Chordia and Shivakumar (2002), and Griffin, Ji and Martin (2003)).

common risk factors play a role in explaining momentum profits and the risk factor is correlated across countries, then country-neutral momentum returns will be correlated in a manner consistent with risk-based explanations. This correlation in momentum returns across markets will also likely have time-varying characteristics to the extent that markets have become more integrated over time.

When testing risk-based explanations of momentum profits, many authors find that the estimated coefficients from regressions of momentum returns on commonly used risk factors are indistinguishable from zero. In fact, the coefficients often have the wrong sign. However, Grundy and Martin (2001) show that momentum portfolios by their very nature have time varying factor coefficients, so a simple examination of the factor loadings of the long and short portfolios is misleading. After controlling explicitly for the time variation in the factor loadings, Grundy and Martin (2001) show that risk based models still do not explain positive average momentum returns.

In another strand of the momentum literature examining risk-based explanations of momentum profits, researchers have investigated the importance of industry effects in momentum. Industry effects are a potential source of common risk, and hence are a good candidate for understanding the source of country-neutral momentum returns and their comovement.⁵ For instance, comovement of industry portfolio returns due to common fundamentals may result in similar industry concentrations in the long and short momentum portfolios across countries, inducing comovement in the country-neutral momentum portfolios. Recent empirical evidence by Grinblatt and Moskowitz (1999) suggests that momentum in industry risk factors explains the profitability of momentum strategies. That is, momentum profits in individual stocks disappear after controlling for industry effects. However, Grundy and Martin (2001) and Asness, Porter, and Stevens (2001) find that the profitability of momentum strategies is not fully explained by industry risk exposure. In particular, Grundy and Martin (2001) find that neither industry

⁵ Industry factors can be viewed as a catch-all risk proxy or an “unpriced” factor. Barberis, Shleifer, and Wurgler (2004), argue that industry factors in returns are at least in part due to industry-level common factors in cash flows, resulting in fundamentals-induced comovement. However, to the extent that investors employ common industry asset allocation strategies across countries, the comovement can also be category-induced. Behavioral underpinnings can also play a role in determining industry sources of momentum effects and their comovement to the extent that there is slow information diffusion within industries and across countries (see Hou (2001), Hong, Lim, and Stein (2000), and Chan, Jegadeesh, and Lakonishok (1996)).

effects nor cross-sectional differences in expected returns explain the level of momentum profits. Rather, the strategies' profitability reflects momentum in the stock-specific (idiosyncratic) component of returns. Asness, Porter, and Stevens (2001) also find both an industry effect and a firm effect independent of industry in momentum.

In contrast to the rational, risk-based explanations, the behavioral models suggest that momentum profits arise from inherent biases in the way that investors interpret information. Barberis, Shleifer, and Vishny (1998), Daniel, Hirshleifer, and Subrahmanyam (1998), and Hong and Stein (1999), develop behavioral models with psychological underpinnings to explain momentum profits. Each of these models addresses particular constraints on investor rationality that cause an under-reaction of prices to information in the short-run, thereby producing return persistence. Brav and Heaton (2002) distinguish between behavioral models and 'rational structural uncertainty' models where investors are rational but have imperfect information, and they find that each class of model has similar implications for asset pricing and therefore provides an 'explanation' for the momentum effect.

While the behavioral models attempt to explain the level of momentum profits, they do not address implications for the comovement of country-neutral momentum strategies. In order for the comovement evidence to be consistent with the behavioral models, the information about firm value to which investors underreact must be correlated across markets. On the other hand, if momentum strategy returns are compensation for some systematic risk, then to the extent that markets are integrated, we would expect to see cross-country comovement in the payoffs to country-neutral momentum strategies.

III. Methodology and Data

A. Methodology

We construct monthly relative strength momentum portfolios by ranking stocks within each country on their total return during the previous 12 months, omitting the month immediately prior. We refer to this variable as $Past(2,12)$. Several recent papers find that skipping a month between portfolio ranking and investment periods avoids confounding bid/ask bounce and nonsynchronous trading microstructure effects. We

define winning and losing stocks as those in the top and bottom three deciles, respectively. The self-financing momentum portfolio return is the return on an equally-weighted portfolio of winners less the equally-weighted portfolio of losers. For each portfolio strategy, we examine the average monthly profit from a strategy of buying a portfolio of winners and shorting a portfolio of losers and holding the position for 1 month.⁶ The portfolios are rebalanced monthly. Although our results are reported in local currency returns, U.S. dollar returns are similar.

In our analysis, we first examine the profitability of unadjusted, industry-adjusted, and risk-adjusted momentum returns across both developed and emerging markets. We then examine the cross-country comovement of momentum returns associated with each of these components. To the extent that there are industry effects in momentum and those industries are significantly related across countries, the correlation in country neutral momentum returns could arise from across-country industry effects in momentum returns. While the industry adjustments can potentially provide some insight into the comovement of momentum returns, they do not necessarily address the underlying source of the comovement, which could plausibly be some common risk factors across markets or a combination of common priced and unpriced factors.

B. Sample Construction

Our main data source is the Thomson Datastream (TDS) database. TDS has return data for more than 30,000 firms in over 50 countries. Our initial sample includes all firms covered by TDS from January 1975 through December 2002. Following Ince and Porter (2004) who show that the TDS must be carefully screened, we screen our sample for issues other than common equity and for firms that are not traded on the country's major exchange(s).⁷ To eliminate double counting and to isolate country

⁶ As further robustness checks, we also employed Fama MacBeth regression techniques in place of momentum portfolio sorts, defined winners and losers as the top and bottom 10%, used value-weighting, and employed alternative prior month return ranking periods and alternative holding periods. In each case, we obtained similar results.

⁷ For example, in the U.S., we include only firms that trade on the NYSE, Amex, or Nasdaq and exclude such non-common equity issues as preferred stock, closed end funds, real estate trusts and investment companies.

effects, we also exclude all listings other than those on a firm's home country exchange (i.e., cross-listings and depository receipts).⁸

During the momentum portfolio formation period, we also exclude from our portfolios any firm whose market capitalization is below the 25th percentile market capitalization of all NYSE stocks, converted to the local currency, in any month during the ranking period. These firms are excluded to avoid significant liquidity constraints, to mitigate any survivorship bias problems, and as Ince and Porter (2004) demonstrate, because many of the difficulties encountered with TDS are concentrated in the smallest deciles. However, our reported conclusions are robust to alternative capitalization screens.

Emerging markets often report erroneous return data. Therefore, following Rouwenhorst (1999), we assign as missing returns below the 2.5 percentile and above the 97.5 percentile of the return distribution for each emerging market in each month during the portfolio formation period. Finally, to insure that we have a series of sufficient length to examine for evidence of comovement, we eliminate countries from the sample that have fewer than 30 valid momentum return months. Our momentum screens eliminate eleven emerging market countries from the sample, leaving a final sample of 16,121 firms from 100 industries and 38 countries over the 1975-2002 time period.⁹

A major strength of TDS is that they use the same criteria for defining industries across countries. As argued by Griffin and Stulz (2001), this consistent classification minimizes the risk of finding low cross-country industry comovement because of firm misclassification. Griffin and Karolyi (1998) also argue that using broad industrial classifications leads to lumping together heterogeneous industries that often include widely disparate lines of business, which can mask the importance of industry effects. In

⁸ Cross-listings and depository receipts can induce a positive dependence in country-neutral momentum returns. For instance, Karolyi and Stulz (1996) find significant return comovements between depository receipts and share returns of the underlying security in the home market as well as aggregate market returns in the home market.

⁹ Following the momentum screens, the eleven emerging market countries dropped from the sample because they have less than 30 valid momentum return months are: Brazil, Columbia, Cyprus, Czech Republic, Egypt, Hungary, Morocco, Pakistan, Peru, Poland, and Venezuela.

our investigation, we use TDS's most disaggregated industry classification, level 6, which includes approximately 100 separate industry definitions.¹⁰

C. Summary Statistics

Table I reports summary statistics for our sample. There are 22 developed and 16 emerging market countries in our sample. Looking at the second column of Table I, we see that most of the start dates for the developed market countries begin in 1975, while the majority of the emerging markets begin in the late 1980s and early 1990s. Not surprisingly, the U.S. and Japan have the largest number of firms and industries, and the largest market capitalizations of all the countries in our sample. Although much smaller on average, over one third of the emerging market countries have both more firms and industries than the bottom third of the developed market countries.

On a country-by-country basis, our reported number of firms and market capitalizations are similar to those reported by Rouwenhorst (1998, 1999) and Chui, Titman, and Wei (2000) who use alternative data sources. Griffin, Ji, and Martin (2003) also use the TDS data, though they do not impose an exchange listing screen, a market capitalization screen, a home country listing screen, an extreme outlier screen, or a valid number of momentum return months screen. However, they delete from their sample countries that have fewer than 50 stocks.

IV. Cross-Country Evidence on the Profitability of Momentum Returns

A. Profitability of Unadjusted Momentum Returns across Countries

The first three columns of Table II report unadjusted momentum profits for portfolios formed on within country $Past(2,12)$ and held for 1 month. We report the average monthly return, t-statistic, and number of months with valid momentum returns, where a valid month is one in which each of the long and short portfolios has at least three stocks. In the next set of columns we report the difference in return between the

¹⁰ As a robustness check, we compare our industry results from the U.S. using the CRSP database with firms sorted into industries by their 4 digit SIC classification and using TDS's level six industry definitions. We obtain very similar results using either dataset. As a further robustness check, we also compare our results for individual countries with those from Rouwenhorst (1998, 1999) who uses a different data source. Again, the results are similar.

portfolio of past winners and the local market return as well as the difference between the local market return and the portfolio of past losers.

The results in Table II show that momentum profits exist across a broad range of countries, though they are not necessarily universal. We find that 14 of 22 developed market countries have a significant momentum effect (13 at the 5% significance level), while 4 of 16 emerging market countries have a significant momentum effect. Interestingly, for Japan, the second largest market, we do not find a significant momentum effect. Chui, Titman, and Wei (2000) and Griffin, Ji, and Martin (2003) also find an insignificant momentum effect for Japan.¹¹ Similarly, for several of the other Asian countries and for the Scandinavian countries, we also do not find a significant momentum effect. For the emerging markets, we find that the proportion of countries that have a significant momentum effect is substantially lower than for the developed markets, but the magnitude of the effect is larger on average, consistent with the smaller number of firms and the higher return volatility in these markets. Recall that we eliminated all stocks below the NYSE 25th percentile, and since both winner and loser portfolios are often concentrated in smaller, higher volatility stocks, this might bias our sample against finding significant momentum profits. However, our results are consistent with those reported by Jegadeesh and Titman (1993, 2001), Rouwenhorst (1998, 1999), Chui, Titman, and Wei (2000), and Griffin, Ji, and Martin (2003) who also find similar momentum returns.

Looking further at the developed versus emerging market momentum returns, we find that the momentum strategy earns an average of 62 basis points per month for the developed markets and 92 basis points per month for the emerging markets over our full sample. The difference in the momentum profits between developed and emerging markets is due to the differences in returns from the short portfolios. The difference in return between the winner portfolio and the local market return is 28 basis points per month for the developed markets and 30 basis points for the emerging markets. However, the difference between the loser portfolio and the local market is 34 basis points for the developed markets and 62 basis points for the emerging markets. The

¹¹ Though not reported, if we exclude the two largest countries (Japan and the U.S.) from our sample, both the magnitude and significance level of the average, country-neutral momentum effect increases.

result from the emerging markets that the returns from the loser portfolio exceeds the winning portfolio on average is consistent with Hong and Stein's (1999) model, which implies that losers losing should drive momentum. However, it is relatively difficult to short stocks in many of the emerging markets.¹² Finally, the results reported at the bottom of each panel also show that there is substantial time variation in the magnitude of the momentum effect across both developed and emerging market countries.

B. Industry and Firm Effects in Momentum Returns

Firms within an industry may have similar exposures to sources of risk, including both priced and unpriced factors. The extent to which industry effects are present in momentum returns is particularly important because they can be a potential source of comovement of country-neutral momentum returns. That is, to the extent that there are industry effects in momentum returns, that there are similar industry concentrations in both long and short momentum portfolios across countries, and that industry returns significantly comove across countries, then country-neutral momentum returns would be correlated.

We create country-neutral industry momentum portfolios and industry-adjusted momentum portfolios by first creating equally-weighted industry portfolios within each country using level 6 TDS industry classifications. Each industry portfolio is required to have a minimum of three firms. We also create industry-adjusted returns by taking the difference between the firm's monthly return and the return on its country-specific industry portfolio. Across-industry momentum portfolios are formed by ranking the total return to each industry portfolio over Past(2,12) and forming an equally-weighted portfolio of the top 30% of industries within the country and shorting the bottom 30% and holding for one month before rebalancing. Within-industry portfolios are formed by ranking total industry-adjusted returns over Past(2,12) and going long the top 30% and short the bottom 30%.

The last six columns of Table II provide returns to both across-industry and within-industry (i.e., industry-adjusted) momentum portfolios. For the across-industry

¹² As shown by Bris, Goetzmann, and Zhu (2003), short selling is difficult to implement and not feasible in many markets.

portfolios in developed countries, we find significant across-industry momentum for over half (12 of 19) of the markets with sufficient data to calculate industry momentum. Of these 12 countries, 10 also have a significant average unadjusted momentum return. Interestingly, for Japan and Switzerland, we find a significant industry momentum effect but no unadjusted momentum effect. We also find that the magnitude of the industry effect in momentum is larger on average than the market-wide momentum effect across the developed countries (i.e., 0.82 for across-industry versus 0.62 market-wide).

For the emerging markets, industry momentum can only be calculated for 9 countries, and it is never significant. However, the number of industries in each market and the average number of firms in each industry portfolio is much smaller for the emerging markets than for the developed markets. Overall, the across-industry momentum effect is nearly zero for the emerging markets.

Similar to the unadjusted market-wide momentum results reported earlier, there is also substantial time series variation in the magnitude of the industry momentum effect. For the developed markets, the industry momentum effect is significant in each sub-period and is largest during the 1996-2002 sample period. The emerging markets, on the other hand, do not display an across-industry momentum effect in the full sample period or in any sub-period.

The last three columns of Table II provide evidence on within-industry (i.e., industry-adjusted) momentum. For the developed markets, 9 of 21 countries have a significant within-industry momentum effect. Of these 9 countries, 7 have both significant across-industry and unadjusted momentum, 1 has unadjusted but no across-industry, and 1 has neither a significant across-industry nor unadjusted momentum effect. For the emerging markets, 3 out of 13 countries have a significantly positive within-industry momentum effect, all of which also display significant unadjusted momentum returns. Taken together, the across-industry and within-industry results show that industry momentum does not subsume unadjusted momentum, consistent with the evidence in Grundy and Martin (2001) and Asness, Porter and Stevens (2001).

We also examined alternative methods for constructing industry portfolios. In addition to value-weighting rather than equally-weighting, we also examined global industry portfolios and a mix of local and global industry portfolios for those industries

where the primary finished product is tradable. See Griffin and Karolyi (1998) for more on the role of traded vs. non-traded industries. Our reported results are robust to each of the alternative methods.

C. Risk Adjusted Momentum Returns

In this section, we document the inability of simple factor models to explain the level of average momentum returns. We use a single-factor market model and a three-factor model similar to that of Fama and French (1992, 1993, 1996). We use two variations of each of these models, one with country-specific versions of the factors and another with global versions. For the country-specific models, our measure of the market return is the value weighted average of all stocks' local currency return. Our size factor, SMB, is calculated using NYSE 30th and 70th percentile breakpoints converted to the local currency.¹³ The Small and Big portfolios are value-weighted and must each have at least three stocks. The SMB factor is the simple difference between the two portfolios. The HML factor is from Ken French's website.

Global versions of the factors are calculated from local values. The global market return is the weighted average of the local market dollar returns using beginning of month market capitalization in U.S. dollars as weights. The global market return is then converted to each local currency. The global SMB factor value-weights the country specific 'Small' and 'Big' portfolios, calculates the difference, and then converts to each local currency. The global HML factor is from Ken French's website.

Since Grundy and Martin (2001) show that the factor coefficients are time varying, we estimate momentum portfolio factor coefficients as follows. For each month and for each stock, we estimate each factor model using a minimum of 24 and a maximum of 36 past months of stock returns and factor realizations to estimate stock specific factor loadings. The momentum portfolio factor loading for month t is then calculated from the individual stock factor loadings calculated from $t-1$ through $t-36$ (i.e., it is the simple average of the factor loadings of the individual stocks). The fitted values

¹³ The sample of stocks used to calculate SMB is not subject to the 25th NYSE market capitalization decile screen as is the sample of stocks from which momentum portfolios are selected.

for time t are then calculated using the estimated portfolio factor loadings and the factor realizations.

Table III provides the risk-adjusted momentum return results. Since we require that a stock must have a minimum of 24 months of prior returns to calculate its factor loadings, the sample used to form momentum portfolios is a bit smaller here than in the earlier reported results in Table II. For comparative purposes, the first column of Table III reports the average unadjusted momentum returns from using this smaller sample. In Panel A, under Model 1, we provide the results from using the country-specific market model, while under Model 2 we provide the Fama-French three-factor model results.¹⁴

Using the local market model, Model 1, we see that the model does a poor job of explaining the level of momentum returns. Of the 13 developed markets with significant momentum returns, the model explains a significant portion for only 2 of the countries. Interestingly, the market model betas are positive and significant for only 2 of the 13 developed markets with significant momentum and 6 have the wrong sign. These results are consistent with Grundy and Martin (2001) who show that a single factor model does a poor job of explaining the level of momentum returns.

Using a local version of the Fama-French three-factor model, our Model 2, we again find that the model does a poor job of explaining the level of momentum returns. The fitted values are positive and significant in only 3 of the developed countries. Of the 13 developed markets with significant momentum returns, we find that 10 have significant average factor loadings on one or more of the factors, though their magnitudes are too low to explain the level of momentum profits. In fact, 6 of the 14 significant coefficients are negative.

In the bottom part of Panel A, we provide the country-specific risk adjustment results for the emerging markets. Similar to the developed market results, we again find that the country-level single and multi-factor specifications do a poor job at explaining the average momentum returns across the emerging markets. Of the two countries with significant momentum returns, neither model is able to explain the level of average returns. In fact, the only significant factor loading has the wrong sign. Four additional

¹⁴ For countries that do not have an HML factor available, we use a two factor model.

countries have average unadjusted momentum returns above 1 percent per month, but the models also do a poor job of explaining these returns.

Panel B of Table III provides the risk-adjusted momentum returns from using global factors in place of the country-specific factors. The results show that the global factor models do a poorer job at explaining the level of country-neutral momentum returns. This is consistent with Griffin (2002) who finds that domestic versions of the Fama-French model have lower pricing errors than do global versions.

V. Cross-Country Comovement of Momentum Returns

The previous section shows that the momentum effect exists for a broad cross-section of countries and has a time varying component, there are both industry and firm effects independent of industry in momentum returns, and standard risk factors do a poor job of explaining average momentum returns.¹⁵ In this section, we examine the cross-country comovement of momentum returns.

A. Correlation of Unadjusted Momentum Returns across Countries

Table IV provides evidence that country-neutral momentum returns significantly comove. Panel A of Table IV reports the average correlation of country-neutral momentum returns within developed markets over the full sample period 1975-2002, while Panel B reports the average correlation within emerging markets over the 1990-2002 sample period. The begin date for the emerging markets correlation results in Panel B corresponds to the starting date available for many of the emerging markets. We also report the average correlation between equally-weighted country market indices for comparison.

For each developed market in Panel A, we find that the average pairwise correlations of country-neutral momentum returns with the other developed markets are significant. The average of the pairwise correlations range from above 0.30 for the U.S., U.K., and France to 0.12 for Austria. With the exception of two of the smaller markets, Austria and New Zealand, we find that for each country, over half of the pairwise

¹⁵ In unreported results, we also find that the factor models do a poor job of explaining the level of across-industry and within-industry momentum.

correlations are individually significant.¹⁶ France has the highest number of positive and significant pairwise correlations at 21, while Austria has the lowest number with 7. No pair of developed countries has a significant negative correlation in their momentum returns.

The range of statistically significant pairwise correlations for the developed markets in seventh column shows that the averages often mask the magnitude of the relationship between country-neutral momentum returns. The momentum returns for the U.K. and the U.S., for instance, have a pairwise correlation of 0.53 over the full sample period. Many of the other developed markets also have substantial pairwise momentum return correlations. In fact, the momentum returns for half of the developed markets have a significant pairwise correlation of 0.40 or greater with another country's momentum returns. At the bottom of Panel A, we also see that the comovement in momentum has grown over time. If we broaden our correlation analysis to the intraregional and interregional levels and mix the developed and emerging markets, we obtain much smaller correlations similar to those reported in Griffin, Ji, and Martin (2003).

For comparison purposes, we also provide the correlation of country return indices. However, it is important to note that the correlation of country return indices does not imply that country-neutral momentum returns should necessarily be correlated. Not surprisingly, the contemporaneous market return average correlations are all positive and significant. The market return correlations are also larger than the average momentum return correlations.

Panel B of Table IV provides the average pairwise correlations for the emerging markets. Relative to the developed market results, the magnitude and significance of the momentum return correlations are substantially smaller. Of the 16 emerging markets, the average correlations of each with the other emerging markets range from -0.07 to 0.06 , with only one of the average correlations positive and significant and one negative and

¹⁶ The correlations for the long portfolios across countries and short portfolios across countries are also similar, suggesting that contagion is not the source of the dependence in momentum returns across markets. Cross country contagion effects can also potentially be a source of the comovement in momentum returns to the extent that momentum profits arise primarily from the relation in short positions across countries. Contagion strategies entail selling assets from one country when crisis hits another (see, for instance, Bekaert, Harvey and Ng (2004) and Kaminsky, Lyons, and Schmukler (2001)).

significant. The average market index correlations are all significantly positive, though lower in magnitude than within developed markets.¹⁷

Panel C of Table IV shows average correlations of developed with emerging markets. The average correlation is positive and significant, though smaller in magnitude than the average within only developed countries.

Taken together, we find that while emerging markets display higher average momentum returns than developed markets, country-neutral momentum returns within emerging markets do not comove. Comovement in country-neutral momentum returns is concentrated primarily among developed countries. Although not conclusive, this pattern is consistent with integrated markets across developed countries but not across emerging countries, and the returns to momentum portfolios being due to an omitted risk factor.

B. Correlation of Industry and Industry-Adjusted Momentum Returns across Countries

If country-specific industry portfolio returns are correlated across countries, then industry effects might explain the comovement of country-neutral momentum returns. For example, if the auto industry has a good year in both the U.S. and Japan, then momentum portfolios in both countries would be long auto industry stocks. To the extent that cross-country industry comovement explains the comovement in unadjusted country-neutral momentum, we would expect country-neutral across-industry momentum to comove and within-industry momentum portfolios not to comove.

Table V examines the extent to which industry and industry-adjusted momentum returns comove. In particular, we provide the average correlation between country pairs for the industry and industry-adjusted momentum returns as well as summary information for all country pair correlations. Panel A of Table V reports the correlations for the developed markets over the full sample period 1975-2002, Panel B reports the correlations for the emerging markets over the 1990-2002 sample period, and Panel C provides summary information on the average pairwise correlations for the industry and industry-adjusted momentum returns between emerging and developed markets.

¹⁷ This is consistent with the findings of others. See, for instance, Goetzmann, Li, and Rouwenhorst (2002) and Longin and Solnik (1995, 2001).

Panel A of Table V reports the average pairwise correlation between developed markets for countries where we can calculate across-industry momentum returns in at least 30 months of the sample period. Although from Table II we see that the level of across-industry momentum (0.82%/month) is higher than unadjusted momentum (0.62%/month), the average developed market across-industry pairwise correlation (0.10) is lower than the unadjusted momentum correlation (0.22). Interestingly, the country-neutral within-industry momentum portfolios are as correlated as the across-industry momentum portfolios. It is clear from Table V that industry comovement does not by itself explain the comovement in momentum returns.

Panel B repeats for emerging markets the analysis of Panel A. Although we find little evidence of comovement of unadjusted momentum returns within emerging markets, across-industry momentum does weakly covary. The average within-emerging pairwise correlation is significant for 3 of 6 emerging markets, although only one individual country pair correlation is significant. The within-industry portfolios show little evidence of comovement, though there is limited industry coverage in the emerging markets.

VI. Risk Factor and Momentum Return Comovement

Industry comovement does not explain the comovement in country-neutral momentum returns. However, it is possible that common asset pricing models, while unable to explain the level of momentum returns, may be able to explain the comovement of momentum returns. To see why, recall from Grundy and Martin (2001) that “buying recent winners and selling recent losers *guarantees* time-varying factor exposures in accordance with the performance of common risk factors during the ranking period.” If risk factors covary, then the time varying factor exposures of the long and short portfolios must also covary.

A. Correlation of Risk Adjusted Momentum Returns across Countries

In Table VI we provide risk-adjusted correlations corresponding to the country-specific versions of the factor models. In Panel A, we use the domestic version of the market model, while in Panel B we use the domestic version of the Fama-French model.

For this analysis, we only include countries that have data available to estimate both models so that we can more accurately compare across models without deviations caused by different countries or sample periods.¹⁸

Looking first at Panel A of Table VI, we find that the correlations of the fitted momentum returns from using the domestic market model are large and significant. The average correlation across countries of these fitted values is 0.36, with a t-statistic of 8.06. In contrast, while the correlations of the errors from the domestic market model are also relatively large and significant, they are less than half as big as the average fitted correlations across countries (0.36 for the average fitted correlations versus 0.17 for the average error correlations). However, we must be careful in comparing the correlations of the fitted values and of the residuals as the variance of the errors is typically much higher, potentially masking differences in covariance. The last three columns decompose the pairwise covariance of country-neutral momentum as:

$$\begin{aligned}\text{Cov}(R_A, R_B) &= \text{Cov}(R_A^{\text{Fit}} + R_A^{\text{Err}}, R_B^{\text{Fit}} + R_B^{\text{Err}}) \\ &= \text{Cov}(R_A^{\text{Fit}}, R_B^{\text{Fit}}) + \text{Cov}(R_A^{\text{Err}}, R_B^{\text{Err}}) + \\ &\quad [\text{Cov}(R_A^{\text{Fit}}, R_B^{\text{Err}}) + \text{Cov}(R_A^{\text{Err}}, R_B^{\text{Fit}})]\end{aligned}$$

$$1 = \text{Percent Fit} + \text{Percent Error} + \text{Percent Other}$$

where R_A is the country-neutral momentum of country A and R_A^{Fit} is the portion of the return explained by the model, and R_A^{Err} is the portion not explained by the model.

On average, the fitted values explain a smaller portion of the covariance relative to the errors (0.29 versus 0.57). The same is true of 17 of 19 developed markets. The sub-sample results at the bottom of the panel further show that there is considerable time variation in the fitted and error correlations and also confirms that the fitted values explain a smaller portion of the covariance relative to the errors through time.

¹⁸ If we allow the sample to vary across models, we obtain qualitatively similar results for the developed countries. Due to insufficient data, many of the models cannot be estimated for the emerging markets, and the fit is generally poorer when the estimates can be obtained.

In Panel B of Table VI, we report the results from using the domestic version of the Fama-French model. The percent of the covariance explained by the fitted values rises from 0.29 to 0.40 when using the three-factor model and the proportion explained by the errors falls from 0.57 to 0.40. On average, the fitted values from using the three-factor model explain an equal proportion of the covariance relative to the errors (0.40 each). However, there is considerable across country variability in this result, with the fitted values for almost half of the countries explaining a higher percentage of the covariance than to the errors. Similar to the single-factor model, the three-factor subsample results at the bottom of the panel also show that there is considerable time variation in the fitted and error correlations. The proportion of the covariance explained by the fitted values and errors also varies considerably over time, with the fitted values explaining an increasing portion of the covariance through time.

We also examined global versions of the factor models. Though not reported, the global versions provide qualitatively similar results. However, the percent of the covariance explained by the errors rises markedly with the global versions, suggesting a general deterioration in the fit of these models relative to the domestic versions. These results are not surprising given our earlier discussion on the relatively poorer fit of the global models relative to the domestic versions as reported by Griffin (2002).

B. Does the Comovement of Risk Factors across Countries Explain the Comovement of Country-Neutral Momentum Returns Across Countries?

The risk-adjusted momentum results of Table VI suggest that there is a relationship between the correlation of momentum returns and risk factors. To explore this relationship in greater detail, we first examine if cross sectional differences in factor covariances explain country-neutral momentum return covariances. We then examine if the covariance of factor returns during the measurement period predict the covariance of momentum returns in the holding period.

For the first set of tests, we calculate the sample pairwise covariance of country-neutral momentum returns and risk factors. We then regress the covariance of momentum returns on the covariance of risk factors for all pairs of developed countries that meet the momentum screens, have local factors that can be calculated, have a valid

momentum return on or within 12 months of the factor realization, and where the covariances are estimated with a minimum of 30 data points. Panel A in Table VII shows the results from these regressions. The first three columns contain univariate regression results corresponding to each risk factor, while the fourth column contains the multivariate regression results with all three risk factors included. The fifth and sixth columns contain the multivariate results over two equal 14-year sub-periods: 1975-1988 and 1989-2002.

The univariate regressions show that the covariance of each risk factor is significant in explaining the covariance of country-neutral momentum returns. In other words, countries whose factors have high covariance also tend to have high covariance of country-neutral momentum.

When all three factors are included, the market risk factor becomes insignificant, although the remaining factors are significant. Moreover, the proportion of the momentum return covariation explained by the covariation in the risk factors increases from as low as 0.05 to 0.23 in the multivariate specification, further suggesting that factor covariance has some power to explain the covariance of momentum returns. In the last two columns, we also see that the factor covariances do a better job at explaining the covariance of momentum returns in the second sub-sample, which corresponds to the period when momentum return covariances are the largest over the entire sample.

In the second set of tests, we examine whether or not the covariance in factor returns during the measurement period predicts the covariance of momentum returns in the holding period. For each country-neutral momentum strategy and for each local factor realization, we calculate the following deviations from country specific time-series means:

$$\begin{aligned}\varepsilon_{i,t} &= R_{i,t} - \bar{R}_i \\ \gamma_{i,t}^{RMRF} &= RMRF_{i,t} - \overline{RMRF}_i,\end{aligned}$$

where $R_{i,t}$ is the country-neutral momentum return in country i in month t , and $RMRF_{i,t}$ is the return to the excess local market factor in country i in month t . We do similar calculations for SMB and HML. We then calculate the cross product for country i and j :

$$y_{i,j,t} = \varepsilon_{i,t} \cdot \varepsilon_{j,t}$$

$$x_{i,j,t} = \left(\frac{1}{11}\right) \cdot \sum_{k=2}^{12} (\gamma_{i,t-k}^{RMRF} \cdot \gamma_{j,t-k}^{RMRF}).$$

For the regression analysis, we use panel regressions of y on x and include time and country-pair dummies. Panel B in Table VII reports the results from these panel regressions.

The first three columns of Panel B in Table VII report panel regression results for univariate and multivariate specifications using only developed countries, while the last two columns report results for the three-factor specification over the two equal 14 year sub-periods. Looking at the first column, we see that covariance of the market factor returns during the measurement period are significant predictors of the covariance of country-neutral momentum returns during the holding period. Turning to the second and third columns with two and three factors respectively, we further see that for all three factors, covariances of the risk factor returns during the measurement period are significant in predicting covariances of country-neutral momentum returns during the holding period. In the last two columns containing the sub-period results, we further see that the covariance of the risk factor returns during the measurement period play an increasingly significant role in predicting covariances of country-neutral momentum returns during the holding period, with increasing coefficient estimates, significance levels and explanatory power over time.

We also performed several robustness checks. First, we included the intersection of developed with emerging markets in the regression analysis and again found that the covariance of market factor returns during the measurement period are significant in explaining subsequent covariances of country momentum returns, though the explanatory power is lower due to the sparser number of emerging market observations.¹⁹ Second, we included several measures of industry overlap in order to examine the possibility that industry concentration in momentum portfolios drive the cross sectional result. It does not. We also examined five-year sub-periods to further assess if there was a stationarity problem with the covariances in the regression analysis. Though the early five-year

¹⁹ The intersection of emerging markets with emerging markets is very sparse across the multivariate specifications and largely insignificant. The HML factor, for instance, is largely unavailable for the emerging markets.

sample periods are sparse, the time varying significance of the results over each of the five-year periods also suggests that there is not a stationarity problem in the regression analysis. Finally, to assess if there are errors in variables problems in the regression analysis, we also used instrumental variables procedures and again obtained similar results to those reported.

VII. Conclusion

In this paper, we examine the profitability and cross-country comovement of country-neutral momentum returns for both developed and emerging markets. Using relative strength portfolio sorts, we first document the profitability of momentum returns across countries and then decompose those returns by industry effects, firm effects independent of their industry and with single and multifactor models. In addition, we examine the relationship in momentum returns across countries and over time.

We show that country-neutral momentum strategies are profitable on average for both developed and emerging markets, yielding 62 and 92 basis points per month respectively over our sample period extending from 1975 – 2002. This result is consistent with the international evidence presented by Rouwenhorst (1998, 1999) and Griffin, Ji, and Martin (2003). We also find both an industry momentum effect and a firm momentum effect independent of its industry for individual countries and on average across countries using country-neutral momentum strategies. Using a market model and the Fama-French three factor model, we find that over two-thirds of the developed countries display significant exposure to one or more of the risk factors, though the models do a poor job at explaining average momentum returns.

We also find that country-neutral momentum returns are significantly related across countries. This comovement is positive and time varying. Conditioning on industry versus firm momentum effects, we find that industry comovement does not explain the comovement in country-neutral momentum returns. Interestingly, we find that while standard factor models do not explain average momentum returns, they do explain a significant portion of the comovement of country-neutral momentum returns. This finding is of particular importance to the asset allocation, diversification, and risk management strategies of investors who employ international momentum strategies since

countries whose capital markets are more highly integrated are likely to have correlated pay-offs to country-specific momentum portfolios. Taken together, our results suggest that risk factors play an important role in explaining the cross-country comovement of momentum returns.

References

- Asness, Clifford S., R. Burt Porter, and Ross L. Stevens, 2001, Predicting Stock Returns Using Industry-Relative Firm Characteristics, Working Paper, University of Florida.
- Barberis, Nicholas, Andrei Shleifer, and Robert Vishny, 1998, A Model of Investor Sentiment, *Journal of Financial Economics* 49, 307–343.
- Barberis, Nicholas, Andrei Shleifer, and Jeffrey Wurgler, 2004, Comovement, *Journal of Financial Economics*, forthcoming.
- Bekaert, Geert, Campbell Harvey, and Angela Ng, 2004, Market Integration and Contagion, *Journal of Business*, forthcoming.
- Bris, Arturo, William N. Goetzman, and Ning Zhu, 2003, Efficiency and the Bear: Short Sales and Markets Around the World, Working Paper, Yale University.
- Brav, A., and J. B. Heaton, 2002, Competing Theories of Financial Anomalies, *Review of Financial Studies* 15, 575-606
- Carhart, M. M., 1997, On persistence in mutual fund performance, *Journal of Finance*, 52(1), 57–82.
- Chan, Louis K. C., Narasimhan Jegadeesh, and Josef Lakonishok, 1996, Momentum strategies, *Journal of Finance* 51, 1681–1713.
- Chui, Andy C. W., Sheridan Titman, and K. C. John Wei, 2000, Momentum, Legal Systems and Ownership Structure: An Analysis of Asian Stock Market, Working paper, University of Texas.
- Chordia, Tarun, and Lakshmanan Shivakumar, 2002, Momentum, business cycle, and time-varying expected returns, *Journal of Finance* 57, 985-1019
- Conrad, Jennifer, Mustafa Gultekin, and Gautam Kaul, 1997, Profitability of short-term contrarian, portfolio strategies: Implications for market efficiency, *Journal of Business and Economic Statistics* 15, 379-386.
- Conrad, Jennifer, and Gautam Kaul, 1998, An anatomy of trading strategies, *Review of Financial Studies* 11, 489–519.
- Daniel, Kent, David Hirshleifer, and Avanidhar Subrahmanyam, 1998, Investor psychology and security market under- and overreactions, *Journal of Finance* 53, 1839–1886.

- Daniel, Kent, M. Grinblatt, S. Titman, and R. Wermers, 1997, Measuring mutual fund performance with characteristic-based benchmarks, *Journal of Finance* 52, 1217-1218.
- DeBondt, Werner F. M., and Richard H. Thaler, 1985, Does the stock market overreact?, *Journal of Finance* 40, 793–805.
- DeBondt, Werner F. M., and Richard H. Thaler, 1987, Further evidence on investor overreaction and stock market seasonality, *Journal of Finance* 42, 557–581.
- Fama, Eugene F., and Kenneth R. French, 1988, Permanent and temporary components of stock prices, *Journal of Political Economy*.
- Fama, Eugene F., and Kenneth R. French, 1992, The cross-section of expected stock returns, *Journal of Finance* 47, 427-465.
- Fama, Eugene F., and Kenneth R. French, 1993, Common risk factors in the returns on stocks and bonds, *Journal of Financial Economics* 33, 3–56.
- Fama, Eugene F., and Kenneth R. French, 1996, Multifactor explanations of asset pricing anomalies, *Journal of Financial Economics* 51, 55–84.
- Fama, Eugene F., and Kenneth R. French, 1998, Value versus growth: The international evidence, *Journal of Finance* 53, 1975–1999.
- Goetzmann, William, Lingfeng Li, and K. Geert Rouwenhorst, 2002, Long-term global market correlations, Working Paper, Yale University.
- Griffin, John M., and Andrew G. Karolyi, 1998, Another look at the role of industrial structure of markets for international diversification strategies, *Journal of Financial Economics* 36.
- Griffin, J., and R. Stulz, 2001, International competition and exchange rate shocks: a cross-country industry analysis of stock returns, *Review of Financial Studies* 14, 215-241.
- Griffin, John M., 2002, Are the Fama and French factors global or country specific?, *Review of Financial Studies* 15, 783-803.
- Griffin, John M., Xiuqing Ji, and J. Spencer Martin, 2003, Momentum investing and business cycle risk: Evidence from pole to pole, *Journal of Finance* 58, 2515-2547.
- Grinblatt, Mark, and Tobias Moskowitz, 1999, Do industries explain momentum?, *Journal of Finance* 54, 1249-1290.

- Grundy, Bruce D., and J. Spencer Martin, 2001, Understanding the nature of the risks and the source of the rewards to momentum investing, *Review of Financial Studies* 14, 29-78.
- Hong, Harrison, Terence Lim, and Jeremy C. Stein, 2000, Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies, *Journal of Finance* 55, 265–295.
- Hong, Harrison, and Jeremy C. Stein, 1999, A unified theory of underreaction, momentum trading and overreaction in asset markets, *Journal of Finance* 54, 2143–2184.
- Hou, Kewei, 2001, Information diffusion and asymmetric cross-autocorrelations in stock returns, Working Paper, The Ohio State University.
- Ince, Ozgur, and R. Burt Porter, 2004, Individual Equity Return Data from Thomson Datastream: Handle with Care!, Working Paper, University of Florida.
- Jegadeesh, Narasimhan, 1990, Evidence of predictable behavior of security returns, *Journal of Finance* 45, 881-898.
- Jegadeesh, Narasimhan, and Sheridan Titman, 1993, Returns to buying winners and selling losers: Implications for stock market efficiency, *Journal of Finance* 48, 65–91.
- Jegadeesh, Narasimhan, and Sheridan Titman, 1995, Overreaction, delayed reaction, and contrarian profits, *Review of Financial Studies* 8, 973–993.
- Jegadeesh, Narasimhan, and Sheridan Titman, 2001, Profitability of momentum strategies: An evaluation of alternative explanations, *Journal of Finance* 56, 699-720.
- Jegadeesh, Narasimhan, and Sheridan Titman, 2002, Cross-sectional and time-series determinants of momentum returns, *Review of Financial Studies* 15, 143-158.
- Kaminsky, Graciela, Richard K. Lyons, and Sergio Schmukler, 2001, Managers, investors, and crises: Mutual fund strategies in emerging markets, forthcoming in the *Journal of International Economics*.
- Karolyi, G. Andrew, and Rene M. Stulz, 1996, Why do markets move together? An investigation of U.S.-Japan stock return comovements, *Journal of Finance* 51, 951-986.
- Kim, Myung Jig, Charles R. Nelson, and Richard Startz, 1991, Mean reversion in stock prices? A reappraisal of the empirical evidence, *Review of Economic Studies* 58, 515-528.

- Lakonishok, J., R. Shleifer, and R. Vishny, 1994, Contrarian investment, extrapolation, and risk, *Journal of Finance* 49, 1541-1578.
- Lo, Andrew W., and A. Craig MacKinlay, 1990, When are contrarian profits due to stock market-overreaction? *Review of Financial Studies* 3, 175–208.
- Longin, F., Solnik, B., 1995, Is correlation in international equity returns constant: 1960-1990, *Journal of International Money and Finance* 14, 3-26.
- Longin, F., Solnik, B., 2001, Extreme correlation of international equity markets, *Journal of Finance* 56, 649-676.
- Rouwenhorst, K. Geert, 1998, International momentum strategies, *Journal of Finance* 53, 267–284.
- Rouwenhorst, K. Geert, 1999, Local return factors and turnover in emerging stock markets, *Journal of Finance* 54, 1439-1464.

Table I
Summary Statistics by Country: 1975 – 2002

<u>Developed Markets:</u>	<u>Starting Date</u>	<u>Number of Firms</u>	<u>Number of Industries</u>	<u>Median Total Market Capitalization</u>
Australia	1975:01	315	70	\$88,943
Austria	1975:01	92	36	\$14,833
Belgium	1975:01	85	45	\$28,891
Canada	1975:01	531	85	\$125,415
Denmark	1975:01	104	34	\$12,692
Finland	1987:01	98	48	\$28,250
France	1975:01	564	86	\$129,567
Germany	1975:01	632	82	\$233,966
Hong Kong	1975:01	352	67	\$59,018
Ireland	1975:01	41	28	\$6,356
Italy	1975:01	256	68	\$89,411
Japan	1975:01	2,035	84	\$2,356,900
Luxembourg	1991:03	24	12	\$11,621
Netherlands	1975:01	191	57	\$96,932
New Zealand	1986:01	46	27	\$13,666
Norway	1975:01	154	41	\$12,544
Singapore	1975:01	157	40	\$32,099
Spain	1986:12	154	48	\$121,057
Sweden	1982:01	310	61	\$73,200
Switzerland	1975:01	382	64	\$52,642
UK	1975:01	1,107	90	\$559,556
<u>USA</u>	1975:01	5,921	100	\$2,211,165
All Developed		13,551		\$6,995,376
<u>Emerging Markets:</u>				
Argentina	1993:07	42	21	\$35,899
Chile	1989:07	91	33	\$42,353
Greece	1988:01	213	55	\$8,756
India	1990:01	295	56	\$79,061
Indonesia	1990:04	124	41	\$20,566
Israel	1993:12	45	26	\$18,770
Korea	1980:10	289	64	\$69,111
Malaysia	1975:01	302	64	\$17,610
Mexico	1988:01	139	39	\$72,285
Philippines	1976:11	85	23	\$7,825
Portugal	1988:01	54	24	\$17,163
Russia	1994:01	71	15	\$40,236
South Africa	1990:01	158	52	\$79,697
Taiwan	1987:09	432	55	\$180,439
Thailand	1987:01	150	46	\$22,755
<u>Turkey</u>	1988:01	80	29	\$5,203
All Emerging		2,570		\$194,064
All		16,121		\$7,211,596

Countries are listed as Developed or Emerging based on the International Finance Corporation's (IFC) categorizations. Start date is the first month that a firm meeting all the data requirements outlined in Section III enters the sample. Number of Firms and Industries are the total number of unique firms and industries represented in the sample. Median total market capitalization is the time series median of the total value (in millions of U.S. dollars) of all firms in the sample in a month for a country converted to US dollars at the corresponding end of month exchange rate.

Table II
Momentum Returns for Unadjusted, Across-Industry, and Within-Industry Portfolios:
Portfolios Formed on Past(2,12) Ranking Period and Held for One Month
1975 – 2002

	Unadjusted			Returns Versus Local Market Index				Across-Industry			Within-Industry		
	Average Monthly Return	t-stat	n	Winner Portfolio	t-stat	Loser Portfolio	t-stat	Average Monthly Return	t-stat	n	Average Monthly Return	t-stat	n
<u>Developed Markets:</u>													
Australia	1.06	3.84	323	0.39	2.75	-0.67	-4.04	-0.05	-0.14	240	0.65	2.16	304
Austria	0.04	0.09	169	-0.11	-0.48	-0.15	-0.51	0.52	0.35	24	-0.44	-0.56	56
Belgium	0.69	2.70	308	0.25	2.07	-0.43	-2.53	0.19	0.50	127	0.07	0.19	148
Canada	1.15	4.48	311	0.49	4.04	-0.66	-3.95	1.53	3.97	304	0.62	3.21	307
Denmark	0.49	1.48	256	0.26	1.55	-0.23	-1.03	0.03	0.05	92	1.05	2.18	162
Finland	0.34	0.52	165	0.27	0.96	-0.07	-0.14	-0.27	-0.19	27	-0.81	-0.78	75
France	0.97	3.63	323	0.44	3.50	-0.53	-3.13	1.35	2.81	251	0.63	2.41	301
Germany	0.85	3.74	323	0.39	3.78	-0.46	-2.93	0.69	3.12	323	0.32	2.16	323
Hong Kong	0.74	2.54	323	0.38	2.20	-0.36	-1.93	1.41	3.30	263	0.32	1.16	323
Ireland	1.46	2.48	149	0.37	1.33	-1.09	-2.54				1.36	0.84	18
Italy	1.79	5.48	323	0.87	5.09	-0.92	-4.48	1.26	2.54	192	0.60	1.74	195
Japan	0.09	0.34	323	-0.01	-0.05	-0.09	-0.68	0.45	2.11	323	-0.18	-1.01	323
Luxembourg	1.30	1.68	83	0.84	2.39	-0.46	-0.83				-0.74	-0.19	3
Netherlands	0.93	3.22	323	0.46	3.40	-0.47	-2.54	1.27	2.87	190	0.38	1.35	191
New Zealand	0.49	0.87	121	0.19	0.70	-0.30	-0.78						
Norway	0.57	1.15	204	0.14	0.54	-0.43	-1.20	0.71	0.66	74	-0.20	-0.37	125
Singapore	0.29	0.89	321	0.05	0.29	-0.24	-1.20	0.65	1.43	194	-0.41	-1.27	230
Spain	0.76	3.24	170	0.58	4.13	-0.17	-1.05	0.88	2.15	125	0.61	2.85	168
Sweden	0.37	0.89	239	0.26	1.52	-0.10	-0.35	0.93	2.04	180	-0.02	-0.05	188
Switzerland	0.95	4.46	295	0.48	4.33	-0.47	-3.49	0.53	1.98	201	0.59	2.70	229
United Kingdom	0.89	3.94	323	0.37	3.64	-0.53	-3.76	0.79	3.79	323	0.48	3.32	323
USA	0.76	3.20	323	0.38	4.27	-0.38	-2.31	1.00	5.37	323	0.53	3.81	323
Total Developed:	0.62	3.61	323	0.28	4.25	-0.34	-2.96	0.82	6.04	323	0.29	2.78	323
1975-1980	0.89	3.24	59	0.42	3.06	-0.47	-3.19	0.84	4.04	59	0.53	3.06	59
1981-1985	0.33	1.30	60	0.07	0.64	-0.25	-1.68	0.47	2.17	60	0.12	0.66	60
1986-1990	0.50	1.83	60	0.19	1.38	-0.32	-1.98	0.54	2.35	60	0.21	1.13	60
1991-1995	0.36	1.53	60	0.17	1.75	-0.19	-1.29	0.63	4.75	60	0.14	0.71	60
1996-2002	0.89	1.64	84	0.46	2.57	-0.43	-1.11	1.38	3.19	84	0.42	1.35	84

Table II Continued

	Unadjusted			Returns Versus Local Market Index				Across-Industry			Within-Industry		
	Average Monthly Return	t-stat	n	Winner Portfolio	t-stat	Loser Portfolio	t-stat	Average Monthly Return	t-stat	n	Average Monthly Return	t-stat	n
<u>Emerging Markets:</u>													
Argentina	0.77	1.22	68	0.14	0.46	-0.63	-1.40						
Chile	0.62	1.51	116	0.38	1.59	-0.24	-0.91				1.63	1.85	34
Greece	0.87	1.14	78	0.51	1.36	-0.36	-0.59	0.20	0.06	8	0.21	0.22	33
India	0.66	1.49	143	0.53	2.34	-0.13	-0.39	0.57	0.80	113	1.17	1.93	118
Indonesia	0.59	0.87	82	0.38	1.01	-0.21	-0.41	-2.30	-1.76	23	-0.21	-0.16	31
Israel	0.82	1.36	56	0.78	1.94	-0.04	-0.10						
Korea	0.57	1.13	149	0.24	0.94	-0.33	-0.99	-1.08	-1.77	104	-0.45	-1.13	108
Malaysia	0.35	0.95	164	0.19	0.87	-0.17	-0.63	-0.34	-0.72	111	0.25	0.73	131
Mexico	1.24	2.56	163	0.55	2.10	-0.69	-2.01	1.19	1.08	63	1.20	2.13	83
Philippines	-2.63	-0.65	79	0.61	1.55	3.24	0.81				0.06	0.05	22
Portugal	1.01	1.60	73	0.92	2.84	-0.09	-0.21				1.08	0.85	13
Russia	4.02	2.09	30	3.22	2.39	-0.80	-0.54				15.14	2.22	4
South Africa	1.37	2.85	131	0.75	3.01	-0.63	-2.13	-4.27	-0.94	119	1.01	2.20	120
Taiwan	0.53	1.23	157	0.31	1.42	-0.22	-0.64	0.68	1.35	145	0.37	1.13	148
Thailand	1.50	2.15	118	1.28	3.43	-0.23	-0.48	0.82	0.52	23	1.15	1.60	53
Turkey	1.11	0.63	37	0.44	0.48	-0.67	-0.45						
Total Emerging:	0.92	3.80	189	0.30	1.87	-0.62	-2.62	-0.01	-0.01	170	0.52	2.55	170
1986-1990	1.74	2.13	45	-0.07	-0.13	-1.81	-2.25	-1.01	-0.69	26	-0.67	-1.20	26
1991-1995	0.69	2.84	60	0.46	2.48	-0.23	-0.95	-0.55	-0.84	60	0.17	0.71	60
1996-2002	0.65	2.41	84	0.39	1.98	-0.26	-1.06	0.70	1.19	84	1.13	3.58	84

Table II Continued

	Unadjusted			Returns Versus Local Market Index				Across-Industry			Within-Industry		
	Average Monthly			Winner		Loser		Average Monthly		Valid Months	Average Monthly		n
	Return	t-stat	n	Portfolio	t-stat	Portfolio	t-stat	Return	t-stat		Return	t-stat	
Total All:	0.61	3.78	323	0.28	4.41	-0.33	-3.05	0.77	5.84	323	0.29	2.90	323
1975-1980	0.89	3.24	59	0.42	3.05	-0.47	-3.19	0.84	4.04	59	0.53	3.11	59
1981-1985	0.33	1.29	60	0.08	0.70	-0.25	-1.63	0.47	2.17	60	0.11	0.61	60
1986-1990	0.50	1.84	60	0.15	1.05	-0.35	-2.15	0.53	2.30	60	0.20	1.07	60
1991-1995	0.39	1.75	60	0.21	2.16	-0.18	-1.26	0.48	3.19	60	0.14	0.76	60
1996-2002	0.87	1.70	84	0.48	2.80	-0.38	-1.07	1.33	3.20	84	0.45	1.50	84

We report the average monthly momentum profits for portfolios of winners minus losers using monthly return data from January 1975 through December 2002. An average return of 0.50 is 50 basis points per month. Winners and losers are defined based on total return during the ranking period of month $t-2$ through $t-12$ (Past(2,12)). Unadjusted refers to the ranking of all stocks by average monthly return within a country during the ranking period. Winners and losers are defined as the top and bottom three deciles, respectively. Across-Industry Portfolios are formed based on total return to equal-weighted industry portfolios during the ranking period. Winning industries are the top 30% and losers the bottom 30% of industries with valid data during the ranking period. The average monthly profit from a strategy of buying an equally weighted portfolio of the winning industry portfolios and shorting the losing industries and holding for 1 month is reported. Within-Industry refers to the ranking of all stocks by difference in return from their equal-weighted industry average. Winners are defined as the top 30% of stocks during the ranking period and losers the bottom 30%. The average monthly profit from a strategy of buying an equal-weighted portfolio of winners and shorting a equal-weighted portfolio of losers and holding the position for 1 month is reported. To be included in any momentum portfolio, a firm must be above the 25th percentile market capitalization of all NYSE stocks in every month during the ranking period. Market capitalizations are converted to dollars, but returns are in local currency. Industry portfolios have a minimum of three firms. Across-industry strategies must have a minimum of four valid industries from which winning and losing industries are selected.

Table III
Risk-adjusted Momentum Returns: Portfolios Formed on Past(2,12) Ranking Period and Held for One Month

Panel A: Risk-adjusted Momentum Returns using Country-specific Market Model and Country-specific Fama-French Three-Factor Model (1975-2002)

<u>Developed Mkts.:</u>	Unadjusted Momentum Returns			Model 1						Model 2									
	Avg	t-stat	n	Fitted	t-stat	Errors	t-stat	B _{MKT}	t-stat	Fitted	t-stat	Errors	t-stat	B _{MKT}	t-stat	B _{SMB}	t-stat	B _{HML}	t-stat
Australia	1.04	3.60	311	-0.06	-0.45	1.10	4.34	-0.03	-1.85	0.30	1.51	0.72	2.65	0.01	0.34	0.03	1.19	0.09	4.03
Austria	-0.05	-0.13	158	0.11	0.46	-0.16	-0.43	0.03	0.87	0.00	0.00	-0.17	-0.30	-.02	-0.31	-0.07	-1.92	0.13	4.39
Belgium	0.74	2.70	292	0.10	1.27	0.64	2.46	0.00	0.22	0.56	2.68	0.36	1.17	-0.03	-1.04	-0.01	-0.20	-0.02	-0.99
Canada	1.14	4.45	307	0.10	0.78	1.04	4.53	0.00	-0.15	0.13	0.86	1.04	4.67	-0.01	-0.47	-0.05	-2.10		
Denmark	0.56	1.67	244	0.22	1.44	0.34	1.05	0.03	0.70	0.24	0.78	0.12	0.27	0.03	0.52	-0.04	-0.98	0.02	0.59
Finland	0.45	0.63	152	0.54	1.64	-0.09	-0.13	0.05	1.65	-0.71	-0.42	1.18	0.76	-0.29	-3.76	-0.19	-2.79	0.09	1.43
France	0.91	3.31	311	0.00	0.02	0.90	3.63	-0.02	-0.98	0.07	0.35	0.78	3.13	0.00	-0.02	0.02	0.98	0.01	0.54
Germany	0.84	3.88	311	0.27	1.94	0.57	3.58	0.04	1.65	0.15	0.87	0.61	3.89	0.06	2.47	0.00	-0.23	0.01	0.44
Hong Kong	0.80	2.71	311	0.23	1.15	0.58	2.18	0.08	4.24	0.13	0.55	0.61	1.95	0.08	4.21	-0.02	-0.76	0.09	4.99
Ireland	1.45	2.26	138	0.14	0.80	1.31	2.12	0.04	1.32	0.23	0.37	0.71	0.90	0.00	0.13	-0.11	-1.94	-0.12	-4.60
Italy	1.73	5.07	311	0.36	2.18	1.37	4.52	0.00	-0.17	0.36	1.72	0.32	1.12	-0.07	-2.91	-0.03	-1.31	0.09	3.24
Japan	0.04	0.14	311	0.05	0.44	-0.02	-0.07	-0.03	1.37	-0.07	-0.37	0.13	0.64	0.06	2.87	0.02	1.26	0.00	0.19
Luxembourg	1.09	1.30	73	0.09	0.11	1.00	1.35	-0.08	-0.90										
Netherlands	0.94	3.22	311	0.04	0.47	0.90	3.18	-0.02	-1.11	0.20	1.12	0.70	2.63	0.03	1.26	0.05	1.72	-0.05	-2.92
New Zealand	0.65	1.15	111	-0.18	-0.87	0.83	1.47	-0.11	-2.98	-0.03	-0.08	1.32	1.24	0.01	0.31	0.16	1.78	0.08	1.55
Norway	0.40	0.75	199	0.02	0.10	0.38	0.72	-0.01	-0.29	-0.43	-0.42	-0.01	-0.01	-0.09	-0.98	-0.01	-0.13	0.00	-0.04
Singapore	0.30	0.99	307	0.16	1.00	0.14	0.43	0.04	2.08	-0.42	-1.02	0.09	0.20	-0.05	-1.63	0.05	1.66	-0.05	-2.02
Spain	0.73	3.12	163	0.38	3.04	0.35	1.57	0.03	1.19	0.37	2.16	0.30	1.20	0.02	1.20	-0.06	-3.05	0.01	0.85
Sweden	0.33	0.80	227	0.16	0.75	0.17	0.48	-0.04	-1.51	0.33	1.44	0.01	0.04	-0.03	-1.58	-0.04	-1.27	0.06	2.62
Switzerland	0.92	4.05	281	0.06	0.74	0.86	4.02	0.03	1.64	0.41	2.18	0.56	2.06	0.05	1.63	0.01	0.18	0.04	2.47
UK	0.88	3.86	311	0.06	0.78	0.83	3.93	-0.01	-0.63	0.07	0.42	0.69	3.81	0.01	0.50	0.02	0.78	-0.03	-1.48
<u>USA</u>	0.75	3.14	311	-0.01	-0.05	0.75	3.93	0.05	2.22	0.09	0.47	0.66	3.63	0.06	3.03	0.08	3.44	-0.08	-3.15
Total Developed:	0.60	3.54	311	0.05	0.58	0.55	4.14			0.09	0.67	0.50	4.03						
<u>Emerging Mkts.:</u>																			
Argentina	0.33	0.45	54	-0.21	-1.09	0.54	0.73	0.03	1.13	-0.02	-0.07	0.17	0.22	0.02	0.56	0.05	1.37		
Chile	0.59	1.41	116	-0.22	-1.83	0.81	1.96	0.00	0.01	0.33	1.44	0.13	0.30	0.04	1.72	0.03	0.70		
Greece	1.13	1.40	67	0.49	2.04	0.64	1.76	-0.08	-2.57	0.32	0.73	1.67	2.30	-0.23	-11.70	0.24	5.00		
India	0.34	0.72	131	0.20	1.41	0.14	0.28	0.05	4.00	0.31	1.62	0.02	0.05	0.05	3.58	0.01	1.04		
Indonesia	-0.07	-0.09	75	-0.13	-0.52	0.06	0.08	-0.01	-0.25	-0.07	-0.12	-0.50	-0.41	0.12	2.78	0.11	1.83		
Israel	1.26	1.47	33	0.12	0.42	1.13	1.43	-0.03	-0.78	0.46	0.62	0.81	0.52	-0.22	-2.47	0.03	0.47		
Korea	0.35	0.67	142	-0.05	-0.32	0.40	0.75	0.02	1.31	-0.73	-2.55	0.79	1.25	0.05	2.92	0.03	1.12		
Malaysia	0.42	1.16	158	0.38	2.12	0.04	0.12	0.02	0.80	0.29	0.90	-0.09	-0.26	-0.02	-0.93	-0.09	-3.42		
Mexico	0.83	1.62	135	-0.01	-0.07	0.84	1.76	0.05	2.01	0.00	0.01	0.77	1.62	0.02	0.55	-0.09	-2.66		
Philippines	-2.95	-0.60	65	0.65	0.78	-3.60	-0.72	-0.35	-3.06	0.96	1.44	-10.44	-0.87	-0.06	-1.09	-0.13	-1.61		
Portugal	1.26	1.87	65	1.02	3.28	0.24	0.36	0.01	0.15	0.87	2.14	1.14	1.43	-0.21	-3.04	0.13	2.28		
Russia	3.49	1.67	27	0.84	0.38	2.65	1.11	0.09	1.50	-0.91	-1.02	2.57	1.05	-0.28	-4.92	-0.20	-2.01		
South Africa	1.34	2.60	119	0.01	0.14	1.33	2.53	-0.02	-1.76	0.06	0.25	1.28	2.30	0.02	0.92	0.00	-0.04		
Taiwan	0.49	1.13	144	-0.06	-0.38	0.55	1.30	0.05	3.28	0.16	0.64	0.33	0.79	0.03	2.28	0.00	0.25		
Thailand	2.13	3.00	108	0.26	1.05	1.87	2.66	0.00	0.05	0.28	0.45	2.35	2.91	-0.11	-3.21	-0.04	-0.93		
<u>Turkey</u>	-1.46	-0.62	22	0.66	1.94	-2.12	-0.88	0.08	2.54										
Total Emerging:	0.82	1.97	255	0.13	0.72	0.69	1.88			0.26	1.38	-0.21	0.83						

Table III continued

Panel B: Risk-adjusted Momentum Returns using Global Market Model and Global Fama-French Three-Factor Model (1975-2002)

<u>Developed Mkts.:</u>	Unadjusted Momentum Returns			Model 3						Model 4									
	Avg	t-stat	n	Fitted	t-stat	Errors	t-stat	B _{MKT}	t-stat	Fitted	t-stat	Errors	t-stat	B _{MKT}	t-stat	B _{SMB}	t-stat	B _{HML}	t-stat
Australia	1.04	3.60	311	0.03	0.36	1.01	3.69	-0.01	-0.28	-0.14	-1.12	1.16	4.00	-0.01	-0.48	-0.03	-1.41	0.05	1.77
Austria	-0.05	-0.13	158	0.01	0.06	-0.06	-0.15	0.02	0.88	-0.23	-1.07	0.06	0.13	0.02	0.59	-0.03	-0.76	-0.05	-1.09
Belgium	0.74	2.70	292	-0.02	-0.34	0.76	2.78	0.02	1.44	0.01	0.07	0.63	2.32	0.03	2.27	0.01	0.72	0.05	2.27
Canada	1.14	4.45	307	0.07	0.69	1.07	4.54	-0.02	-0.69	-0.15	-0.99	1.28	5.34	-0.02	-0.95	-0.10	-3.75	0.00	0.01
Denmark	0.56	1.67	244	0.01	0.08	0.55	1.68	-0.10	-4.42	-0.17	-1.28	0.70	2.03	-0.10	-4.58	-0.01	-0.50	-0.08	-2.82
Finland	0.45	0.63	152	0.29	1.26	0.16	0.23	-0.05	-1.05	0.45	1.51	-0.21	-0.28	-0.03	-0.71	0.02	0.50	0.33	5.62
France	0.91	3.31	311	0.11	1.16	0.80	3.17	0.04	1.68	-0.14	-0.88	1.01	4.17	0.05	2.23	-0.09	-3.98	0.10	3.09
Germany	0.84	3.88	311	0.09	1.16	0.75	3.90	0.00	0.17	-0.07	-0.69	0.83	4.58	0.01	0.43	-0.05	-2.77	0.01	0.38
Hong Kong	0.80	2.71	311	0.20	1.93	0.60	2.01	0.09	3.63	-0.05	-0.32	0.86	2.53	0.08	3.63	-0.09	-2.35	0.04	0.97
Ireland	1.45	2.26	138	0.13	0.79	1.32	2.14	0.03	0.81	-0.13	-0.46	1.04	1.75	0.03	1.16	-0.11	-1.79	0.06	1.07
Italy	1.73	5.07	311	0.05	0.52	1.68	5.01	0.01	0.63	-0.17	-0.94	1.88	5.07	0.02	0.84	0.01	0.30	0.03	0.81
Japan	0.04	0.14	311	-0.01	-0.07	0.04	0.16	0.01	0.73	-0.17	-1.10	0.24	0.97	0.01	0.73	-0.02	-0.74	0.06	2.01
Luxembourg	1.09	1.30	73	-0.16	-0.42	1.25	1.59	-0.03	-0.39	-0.79	-1.41	1.84	2.16	0.02	0.17	0.04	0.40	0.19	3.16
Netherlands	0.94	3.22	311	0.02	0.25	0.92	3.32	0.01	0.73	-0.14	-0.84	1.04	3.89	0.04	2.66	0.01	0.25	-0.09	-2.27
New Zealand	0.65	1.15	111	-0.02	-0.11	0.67	1.25	-0.13	-2.85	-0.46	-1.66	1.14	2.05	-0.14	-3.00	-0.03	-0.78	-0.17	-3.37
Norway	0.40	0.75	199	-0.10	-0.65	0.49	0.93	-0.02	-0.39	-0.19	-0.80	0.34	0.58	0.01	0.47	0.07	2.09	-0.07	-1.27
Singapore	0.30	0.99	307	0.13	1.15	0.17	0.55	0.02	0.60	0.05	0.34	0.17	0.49	0.01	0.47	-0.01	-0.27	-0.01	-0.21
Spain	0.73	3.12	163	0.15	1.86	0.59	2.49	0.09	4.60	0.04	0.29	0.63	2.46	0.07	3.66	-0.08	-3.34	-0.09	-3.23
Sweden	0.33	0.80	227	0.02	0.12	0.31	0.82	-0.12	-4.54	-0.49	-2.72	0.67	1.89	-0.12	-5.30	-0.10	-3.53	0.10	2.39
Switzerland	0.92	4.05	281	-0.01	-0.10	0.93	4.30	0.00	0.13	-0.12	-1.04	0.92	4.25	0.01	0.54	0.01	0.24	-0.05	-2.22
UK	0.88	3.86	311	0.10	1.18	0.79	3.87	-0.03	-1.59	-0.05	-0.35	0.82	4.28	-0.01	-0.57	0.00	0.03	0.05	1.63
USA	0.75	3.14	311	-0.01	-0.05	0.75	3.83	0.02	0.65	-0.16	-0.95	0.84	4.34	0.02	0.88	0.06	2.60	-0.05	-1.44
Total Developed:	0.60	3.54	311	0.02	0.19	0.58	4.23			-0.15	-1.11	0.70	5.37						
<u>Emerging Mkts.:</u>																			
Argentina	0.33	0.45	54	0.02	0.16	0.31	0.40	-0.21	-4.96	-0.25	-0.72	0.59	0.65	-0.18	-4.13	0.26	1.94	-0.29	-2.07
Chile	0.59	1.41	116	0.07	0.69	0.52	1.21	0.01	0.37	0.40	1.62	0.03	0.05	0.01	0.19	-0.03	-0.71	0.12	2.76
Greece	1.13	1.40	67	0.16	0.93	0.97	1.20	-0.23	-5.85	-0.21	-0.55	0.97	0.94	-0.13	-2.37	0.24	2.09	-0.22	-1.50
India	0.34	0.72	131	-0.08	-0.97	0.41	0.86	0.01	0.68	0.00	0.01	0.25	0.46	0.01	0.28	-0.11	-2.74	0.17	3.60
Indonesia	-0.07	-0.09	75	-0.11	-0.47	0.03	0.04	0.01	0.22	-0.50	-1.76	0.43	0.53	0.03	0.66	0.08	1.23	0.07	0.92
Israel	1.26	1.47	33	0.15	0.98	1.11	1.26	-0.05	-1.67	-0.45	-1.16	1.45	1.37	-0.01	-0.34	0.20	6.94	-0.02	-0.47
Korea	0.35	0.67	142	-0.02	-0.18	0.37	0.69	0.02	1.00	0.27	1.55	-0.04	-0.06	0.04	1.64	0.05	1.53	0.12	2.89
Malaysia	0.42	1.16	158	0.10	0.91	0.32	0.87	-0.01	-0.23	-0.27	-1.33	0.55	1.34	-0.03	-1.02	-0.14	-4.20	-0.18	-4.76
Mexico	0.83	1.62	135	0.02	0.11	0.81	1.69	0.05	1.42	0.13	0.51	0.94	1.72	0.00	0.10	-0.13	-2.99	0.04	1.27
Philippines	-2.95	-0.60	65	-0.97	-1.22	-1.97	-0.39	-0.64	-2.51	1.73	0.92	-4.68	-0.88	-0.07	-0.98	-1.83	-2.50	-3.10	-2.72
Portugal	1.26	1.87	65	0.43	1.61	0.83	1.22	0.05	0.69	0.19	0.68	0.95	1.24	0.10	1.45	0.17	3.71	0.16	2.10
Russia	3.49	1.67	27	0.67	0.60	2.82	1.14	0.74	2.93	1.77	0.56	2.63	0.61	0.34	1.47	-0.67	-1.90	0.50	1.34
South Africa	1.34	2.60	119	-0.03	-0.42	1.37	2.66	-0.02	-0.76	-0.02	-0.09	1.49	2.99	-0.02	-1.14	0.03	0.56	0.26	6.73
Taiwan	0.49	1.13	144	0.14	1.19	0.34	0.82	0.00	-0.13	0.07	0.37	0.29	0.66	0.01	0.21	-0.02	-0.58	0.00	0.11
Thailand	2.13	3.00	108	-0.02	-0.17	2.15	3.13	0.03	0.83	0.40	1.35	1.62	2.17	0.05	1.30	-0.06	-0.97	-0.04	-0.74
Turkey	-1.46	-0.62	22	-0.64	-1.12	-0.82	-0.33	-0.54	-2.31	0.24	0.27	-1.70	-0.69	-0.59	-2.57	0.28	0.92	0.39	0.90
Total Emerging:	0.82	1.97	255	0.58	0.62	0.76	1.85			0.04	0.21	0.74	1.66						

Table III continued

Model 1 – Country-specific market model

Model 2 – Country-specific Fama-French three-factor model

Model 3 – Global market model

Model 4 – Global Fama-French three-factor model

We use both a single-factor market model and a three-factor model similar to that of Fama-French. We use two variations of these models – one with country-specific versions of the factors and another with global versions of the factors. For the country-specific market model, we use the value weighted average of all stocks' local currency return within the country where there are three or more firms. For the Fama-French three factor country-specific model, we use the above market factor along with an SMB factor calculated using NYSE 30/70 breakpoints converted to local currency. Small and Big portfolios are value weighted and must have at least three stocks. The SMB factor is the simple difference between the two portfolios. The HML factor is from Ken French's website. Global versions of the factors are calculated from local values. The global market return is the weighted average of the local market dollar returns using beginning of month market capitalization in U.S. dollars as weights. The global market return is then converted to each local currency. The global SMB factor value-weights the country specific 'Small' and 'Big' portfolios, calculates the difference, then converts to each local currency. The global HML factor is from Ken French's website. Since we require that a stock must have a minimum of 24 months of prior returns to calculate its factor loadings, the sample used to form momentum portfolios is a bit smaller here than the reported results in Table II. For comparative purposes, the first column reports the average unadjusted momentum returns from using this smaller sample.

Table IV
Average Pairwise Correlations of Unadjusted Momentum Returns across Countries:
Portfolios Formed on Past(2,12) Ranking Period and Held for One Month

Panel A: Average Pairwise Correlations of Unadjusted Momentum Returns for Developed Markets (1975-2002)

<u>Developed Markets:</u>	Average of All Correlations	n	t-statistic	Average of Significant Correlations	# of Significant (-) Correlations	# of Significant (+) Correlations	Range of Significant (+) Correlations	Average Local Market Index Correlations	t-statistic
Australia	0.17	21	8.61	0.22	0	13	0.13 - 0.35	0.38	12.78
Austria	0.12	21	7.01	0.20	0	7	0.16 - 0.24	0.37	14.17
Belgium	0.15	21	11.09	0.19	0	11	0.13 - 0.25	0.44	14.29
Canada	0.26	21	9.33	0.30	0	17	0.13 - 0.53	0.43	14.46
Denmark	0.24	21	10.32	0.27	0	18	0.16 - 0.38	0.35	14.51
Finland	0.20	21	7.86	0.27	0	14	0.15 - 0.41	0.48	16.46
France	0.32	21	14.56	0.32	0	21	0.17 - 0.49	0.47	14.58
Germany	0.29	21	9.91	0.31	0	19	0.13 - 0.48	0.48	15.14
Hong Kong	0.14	21	10.26	0.18	0	12	0.12 - 0.26	0.34	12.20
Ireland	0.26	21	12.07	0.29	0	16	0.16 - 0.42	0.10	6.26
Italy	0.20	21	9.91	0.23	0	17	0.12 - 0.41	0.41	14.91
Japan	0.15	21	8.89	0.19	0	12	0.12 - 0.29	0.27	18.55
Luxembourg	0.27	21	9.28	0.38	0	11	0.24 - 0.48	0.45	12.92
Netherlands	0.27	21	11.56	0.28	0	20	0.12 - 0.47	0.51	16.29
New Zealand	0.16	21	10.38	0.24	0	8	0.19 - 0.33	0.35	13.58
Norway	0.17	21	8.42	0.23	0	12	0.19 - 0.31	0.46	18.23
Singapore	0.13	21	7.55	0.18	0	13	0.12 - 0.33	0.33	13.85
Spain	0.16	21	13.45	0.19	0	13	0.15 - 0.25	0.46	13.07
Sweden	0.27	21	8.62	0.33	0	16	0.13 - 0.47	0.46	14.96
Switzerland	0.25	21	14.82	0.25	0	20	0.12 - 0.39	0.54	18.92
United Kingdom	0.32	21	12.81	0.33	0	20	0.19 - 0.53	0.49	21.90
USA	0.31	21	10.19	0.32	0	20	0.13 - 0.53	0.45	15.32
Total Developed:	0.22	231	28.05	0.27	0	165	0.12 – 0.53	0.41	39.58
1975-1980	0.09	91	5.56	0.34	0	10	0.27 – 0.46	0.30	20.95
1981-1985	0.02	89	1.28	0.26	1	4	0.31 – 0.54	0.29	22.69
1986-1990	0.09	120	6.17	0.32	0	20	0.26 – 0.52	0.52	51.55
1991-1995	0.16	210	14.78	0.35	0	53	0.26 – 0.57	0.45	40.61
1996-2002	0.33	231	26.86	0.42	0	154	0.22 – 0.82	0.45	30.06

Table IV continued

Panel B: Average Pairwise Correlations of Unadjusted Momentum Returns for Emerging Markets (1990-2002)

<u>Emerging Markets:</u>	Average of All Correlations	n	t-statistic	Average of Significant Correlations	# of Significant (-) Correlations	# of Significant (+) Correlations	Range of Significant (+) Correlations	Average Local Market Index Correlations	t-statistic
Argentina	0.06	12	1.52	0.27	0	1	0.27 - 0.27	0.25	5.59
Chile	-0.03	14	-1.04		0			0.20	6.26
Greece	-0.01	13	-0.14	-0.02	2	2	0.23 - 0.27	0.13	5.85
India	0.02	15	0.52	0.27	0	1	0.27 - 0.27	0.14	5.19
Indonesia	0.00	12	0.10	0.30	0	1	0.30 - 0.30	0.27	8.47
Israel	0.04	8	0.88		0			0.22	8.01
Korea	0.02	13	0.51	0.27	0	1	0.27 - 0.27	0.12	7.32
Malaysia	0.05	15	1.28	0.23	0	1	0.23 - 0.23	0.22	5.72
Mexico	0.04	15	2.15		0			0.23	5.60
Philippines	-0.07	12	-1.37	-0.09	2	1	0.30 - 0.30	0.14	5.01
Portugal	-0.04	13	-1.56		0			0.17	6.02
Russia	-0.05	6	-2.47		0			0.18	5.52
South Africa	-0.04	15	-1.05	-0.21	2			0.13	6.65
Taiwan	-0.02	15	-0.75	-0.23	1			0.18	6.10
Thailand	0.02	13	0.77	-0.20	1			0.19	6.16
Turkey	0.07	7	1.06		0			0.11	3.72
Total Emerging:	0.00	99	0.20	0.01	4	4	0.23 - 0.30	0.18	15.71
1991-1995	0.00	36	0.06	-0.36	1	0		0.13	8.13
1996-2002	-0.02	89	-1.02	0.10	2	4	0.28 - 0.36	0.24	16.76

Table IV continued**Panel C: Average Pairwise Correlations of Unadjusted Momentum Returns Across Developed and Emerging Markets**

	Average of All Correlations	n	t-statistic	Average of Significant Correlations	# of Significant (-) Correlations	# of Significant (+) Correlations	Range of Significant (+) Correlations	Average Local Market Index Correlations	t-statistic
<u>All</u>									
Emerging w/ Developed	0.07	349	10.04	0.20	7	47	0.16 – 0.48	0.23	31.55
<u>1990-1995</u>									
Emerging w/ Developed	0.03	189	2.81	0.20	3	14	0.26 – 0.46	0.17	16.40
<u>1996-2002</u>									
Emerging w/ Developed	0.09	329	9.27	0.23	7	47	0.22 – 0.54	0.29	36.53

We report correlations of unadjusted momentum returns for portfolios of winners minus losers from 1975-2002 for developed countries in Panel A and from 1990-2002 for emerging markets in Panel B. In Panel C, we report average correlations of unadjusted momentum returns between developed and emerging markets. Winners and losers are defined based on average monthly return during the ranking period. Unadjusted refers to the ranking of all stocks by total return within a country during the ranking period. Winners are defined as the top and bottom 30% of firms. To be included in a portfolio, a firm must be above the 25th percentile market capitalization of all NYSE stocks in a given month. Market capitalizations are converted to dollars, but returns are in local currency.

Table V

**Average Pairwise Correlations of Across-Industry and Within-Industry Momentum Returns across Countries:
Portfolios Formed on Past(2,12) Ranking Period and Held for One Month**

Panel A: Developed Markets (1975-2002)

Pairwise Correlations of Industry and Industry-adjusted Momentum Returns														
Developed Markets:	Across-Industry Momentum							Within-Industry-adjusted Momentum						
	All Correlations			Individually Significant Correlations				All Correlations			Individually Significant Correlations			
	Average	n	t-stat.	Average	N (Negative)	N (Positive)	Range	Average	n	t-stat.	Average	N (Negative)	N (Positive)	Range
Australia	0.07	16	2.36	0.13	1	5	0.15 - 0.27	0.07	18	3.95	0.15	0	4	0.14 - 0.16
Austria								0.05	17	1.16	0.38	0	2	0.35 - 0.40
Belgium	0.02	16	0.91					0.08	18	2.52	0.22	0	6	0.17 - 0.25
Canada	0.18	16	4.50	0.28	0	10	0.14 - 0.48	0.14	18	6.81	0.20	0	9	0.14 - 0.31
Denmark	-0.02	16	-1.09					0.01	18	0.60				
Finland								0.01	17	0.32	0.24	0	1	0.24 - 0.24
France	0.17	16	4.41	0.27	0	9	0.13 - 0.40	0.12	18	4.66	0.20	0	9	0.14 - 0.36
Germany	0.15	16	4.42	0.24	0	9	0.13 - 0.45	0.19	18	6.20	0.27	0	12	0.17 - 0.39
Hong Kong	0.03	16	1.37	0.15	0	3	0.15 - 0.16	0.07	18	3.26	0.19	0	4	0.15 - 0.24
Ireland		0			0									
Italy	0.18	16	3.87	0.31	0	9	0.15 - 0.50	0.17	18	4.93	0.28	0	9	0.16 - 0.39
Japan	0.10	16	3.86	0.21	0	6	0.14 - 0.29	0.12	18	6.40	0.17	0	8	0.13 - 0.24
Luxembourg														
Netherlands	0.15	16	3.31	0.31	0	7	0.20 - 0.45	0.14	18	5.31	0.23	0	9	0.17 - 0.32
New Zealand														
Norway	0.00	16	0.14	-0.23	1			0.02	18	0.72	0.18	0	1	0.18 - 0.18
Singapore	0.00	16	-0.19	-0.23	1			0.14	18	5.06	0.24	0	8	0.16 - 0.40
Spain	0.06	16	2.57	0.19	0	1	0.19 - 0.19	0.10	18	5.28	0.20	0	4	0.16 - 0.24
Sweden	0.04	16	1.76	0.00	1	1	0.18 - 0.18	0.09	18	3.82	0.20	0	6	0.17 - 0.25
Switzerland	0.14	16	3.64	0.22	1	7	0.21 - 0.35	0.13	18	8.59	0.18	0	9	0.14 - 0.24
UK	0.22	16	5.06	0.34	0	9	0.15 - 0.50	0.17	18	5.87	0.25	0	11	0.13 - 0.35
USA	0.19	16	3.81	0.26	1	10	0.16 - 0.48	0.16	18	5.36	0.26	0	10	0.19 - 0.35
Total Developed:	0.10	136	7.56	0.24	3	43	0.13 - 0.50	0.11	170	11.26	0.23	0	61	0.13 - 0.40
1975-1980	0.02	21	0.50	0.47	0	1	0.47 – 0.47	0.09	28	3.43	-0.29	1	0	
1981-1985	0.03	15	0.69	0.41	0	2	0.36 – 0.46	0.04	35	1.24	0.03	2	3	0.28 – 0.34
1986-1990	0.03	78	1.46	0.02	4	5	0.30 – 0.42	0.06	78	4.39	0.34	0	4	0.29 – 0.40
1991-1995	0.05	119	4.48	0.07	2	3	0.30 – 0.34	0.12	153	10.22	0.32	0	32	0.26 – 0.50
1996-2002	0.17	120	8.34	0.36	2	48	0.22 – 0.72	0.16	153	10.99	0.35	1	49	0.22 – 0.56

Table V continued

Panel B: Emerging Markets (1990-2002)

	Pairwise Correlations of Industry and Industry-adjusted Momentum Returns													
	Across-Industry Momentum							Within-Industry-adjusted Momentum						
	All Correlations			Individually Significant Correlations				All Correlations			Individually Significant Correlations			
	Average	n	t-stat.	Average	N (Negative)	N (Positive)	Range	Average	n	t-stat.	Average	N (Negative)	N (Positive)	Range
Emerging Markets:														
Argentina								-0.09	4	-1.79				
Chile								-0.29	1					
Greece								0.01	8	0.52				
India	-0.01	5	-0.14					-0.02	5	-0.22				
Indonesia											-0.40	1		
Israel								0.02	7	0.36				
Korea	0.10	5	2.08					-0.01	7	-0.17				
Malaysia	0.11	5	3.38					0.00	6	-0.03	-0.40	1		
Mexico	0.04	5	0.53								0.25	0	1	0.25 - 0.25
Philippines														
Portugal														
Russia								-0.03	8	-0.60				
South Africa	0.09	5	2.12	0.21		1	0.21 - 0.21	-0.05	9	-1.02				
Taiwan	0.05	5	0.99	0.21		1	0.21 - 0.21	0.02	5	0.31	0.25	0	1	0.25 – 0.25
Thailand														
Turkey								-0.09	4	-1.79				
Total Emerging:	0.06	15	1.98	0.21	0	1	0.21 - 0.21	-0.02	30	-0.73	-0.07	1	1	0.25 – 0.25
1991-1995	0.07	10	1.59	0.27	0	1	0.27 – 0.27	0.00	14	-0.05	0.31	0	1	0.31 – 0.31
1996-2002	0.08	10	1.27	0.38	0	1	0.38 – 0.38	-0.02	15	-0.41	0.33	0	1	0.33 – 0.33

Table V continued

Panel C: Across Developed and Emerging Markets

	Pairwise Correlations of Industry and Industry-adjusted Momentum Returns													
	Across-Industry Momentum							Within-Industry-adjusted Momentum						
	All Correlations			Individually Significant Correlations				All Correlations			Individually Significant Correlations			
	Average	n	t-stat.	Average	N (Negative)	N (Positive)	Range	Average	n	t-stat.	Average	N (Negative)	N (Positive)	Range
<u>All</u> Emerging w/ Developed	0.02	101	1.36	0.03	9	14	0.17 - 0.39	0.01	179	1.08	0.07	4	9	0.20 - 0.49
<u>1990-1995</u> Emerging w/ Developed	0.02	79	1.44	0.32	0	4	0.26 – 0.37	0.00	107	-0.13	0.02	1	1	0.31 – 0.31
<u>1996-2002</u> Emerging w/ Developed	0.05	78	2.11	0.07	8	14	0.22 – 0.44	0.03	121	2.01	0.10	5	11	0.25 – 0.56

We report average correlations of across-industry and within-industry momentum portfolio returns from 1975-2002 for developed countries in Panel A and from 1990-2002 for emerging markets in Panel B. In Panel C, we report average correlations of across-industry and within-industry momentum portfolio returns between developed and emerging markets. Winners and losers are defined based on total return during the ranking period. Across-Industry Portfolios are formed based on total return to equal-weighted industry portfolios during the ranking period. Winning industries are the top 30% and losers the bottom 30% of industries with valid data during the ranking period. The average monthly profit from a strategy of buying an equally weighted portfolio of winning industry portfolios and shorting the losing industries and holding for 1 month is reported. Within-Industry Portfolios refers to the ranking of all stocks by difference in return from their equal-weighted industry average. Winners are defined as the top 30% of stocks during the ranking period and losers the bottom 30%. The average monthly profit from a strategy of buying an equal-weighted portfolio of winners and shorting a equally-weighted portfolio of losers and holding the position for 1 month is reported. To be included in a portfolio a firm must be above the 25th percentile market capitalization of all NYSE stocks in a given month. Market capitalizations are converted to dollars but returns are in local currency. Industry portfolios have a minimum of three firms. Across-industry strategies must have a minimum of four valid industries from which winning and losing industries are selected.

Table VI

**Average Pairwise Correlations of Risk-adjusted Momentum Returns across Countries:
Portfolios Formed on Past(2,12) Ranking Period and Held for One Month (1975 – 2002)**

Panel A: Correlations of Risk-adjusted Momentum Returns using Country-specific Market Model

	Correlations of Unadjusted Momentum Returns							Correlation of Fitted Values							Correlation of Model Residuals							Cov. Decomposition		
	All		Significant Pairs					All		Significant Pairs					All		Significant Pairs					All		
	Avg	t-stat	Avg	t-stat	n	Range		Avg	t-stat	Avg	t-stat	n	Range		Avg	t-stat	Avg	t-stat	n	Range		% Fit	% Err	% Other
Australia	0.18	10.24	0.23	18.52	8	0.18 0.28		0.22	4.22	0.31	4.80	12	-0.25 0.64		0.13	6.56	0.20	14.63	7	0.15 0.26		0.17	0.67	0.15
Austria	0.26	10.42	0.33	19.58	8	0.28 0.40		0.28	4.18	0.40	3.89	9	-0.35 0.64		0.12	5.02	0.27	20.15	2	0.26 0.28		0.34	0.40	0.26
Belgium	0.18	8.75	0.24	12.58	9	0.17 0.34		0.28	9.06	0.33	12.02	14	0.17 0.47		0.15	8.87	0.22	16.81	6	0.18 0.28		0.16	0.67	0.17
Canada																								
Denmark	0.26	8.57	0.32	14.24	13	0.20 0.45		0.43	9.75	0.47	12.26	16	0.24 0.69		0.19	6.92	0.28	14.53	8	0.21 0.38		0.32	0.63	0.05
Finland	0.28	6.63	0.46	19.24	5	0.41 0.55		0.44	10.46	0.55	24.57	11	0.41 0.64		0.13	3.30	0.40		1	0.40 0.40		0.59	0.45	-0.04
France	0.32	12.76	0.34	12.14	15	0.19 0.53		0.48	10.07	0.50	10.81	17	0.14 0.80		0.22	8.16	0.29	11.28	11	0.19 0.46		0.32	0.57	0.11
Germany	0.32	11.17	0.35	13.96	15	0.18 0.51		0.44	7.82	0.52	12.95	15	0.13 0.74		0.20	10.00	0.24	13.25	12	0.13 0.39		0.42	0.46	0.12
Hong Kong	0.18	7.95	0.21	9.33	13	0.12 0.43		0.19	5.82	0.24	8.32	13	0.14 0.50		0.08	4.20	0.15	7.26	4	0.12 0.22		0.20	0.50	0.30
Ireland	0.31	11.91	0.35	13.62	13	0.23 0.51		0.21	4.22	0.28	4.56	10	-0.25 0.45		0.26	9.91	0.35	14.66	9	0.24 0.48		0.10	0.78	0.13
Italy	0.27	9.02	0.32	10.70	13	0.17 0.53		0.40	7.97	0.50	10.99	13	0.24 0.83		0.19	6.25	0.31	10.43	8	0.22 0.46		0.39	0.57	0.05
Japan	0.19	11.13	0.22	15.23	13	0.12 0.29		0.17	4.35	0.26	6.53	10	0.13 0.61		0.13	8.96	0.18	13.81	8	0.12 0.24		0.14	0.54	0.32
Luxembourg																								
Netherlands	0.32	10.84	0.36	13.62	15	0.18 0.55		0.46	11.31	0.46	11.31	18	0.18 0.78		0.23	10.07	0.27	15.48	13	0.15 0.40		0.21	0.64	0.15
New Zealand																								
Norway	0.26	7.92	0.43	13.03	3	0.38 0.49		0.61	13.13	0.68	19.18	14	0.45 0.87		0.14	6.10	0.40		1	0.40 0.40		0.54	0.47	-0.01
Singapore	0.16	6.07	0.27	7.43	6	0.19 0.43		0.16	3.38	0.23	2.99	10	-0.35 0.62		0.11	5.08	0.23	5.71	3	0.18 0.31		0.16	0.55	0.29
Spain	0.17	6.03	0.26	6.59	8	0.17 0.49		0.34	9.84	0.38	12.59	15	0.21 0.67		0.11	3.80	0.27	5.34	4	0.17 0.40		0.34	0.66	0.01
Sweden	0.28	10.10	0.31	11.46	14	0.15 0.44		0.38	9.36	0.41	10.38	16	0.14 0.62		0.19	7.75	0.25	12.51	11	0.16 0.40		0.37	0.59	0.05
Switzerland	0.33	12.06	0.35	12.76	16	0.15 0.51		0.48	10.35	0.48	10.35	18	0.20 0.87		0.22	9.83	0.26	9.51	12	0.15 0.46		0.22	0.58	0.19
UK	0.35	15.53	0.35	14.62	17	0.21 0.51		0.42	8.52	0.48	10.42	15	0.14 0.77		0.24	10.15	0.28	12.22	13	0.14 0.46		0.17	0.59	0.24
USA	0.32	11.80	0.33	11.29	14	0.14 0.48		0.38	9.25	0.42	11.36	16	0.20 0.69		0.22	7.08	0.30	11.97	11	0.20 0.48		0.28	0.57	0.15
Total	0.26	9.94	0.32	13.15				0.36	8.06						0.17	7.26						0.29	0.57	0.14
1975-1980	0.13	2.78	0.40	24.64				-0.03	-0.33						0.10	1.98						0.01	0.69	0.30
1981-1985	0.07	1.15	0.36	15.64				0.07	1.55						0.08	1.40						0.18	0.74	0.08
1986-1990	0.05	1.09	0.37	15.46				0.09	1.13						0.04	0.90						0.15	0.63	0.22
1991-1995	0.21	7.53	0.33	11.87				0.29	6.70						0.15	5.43						0.13	0.60	0.26
1996-2002	0.33	9.02	0.42	13.60				0.41	8.19						0.22	6.65						0.30	0.56	0.15

Table VI continued

Panel B: Correlations of Risk-adjusted Momentum Returns using Country-specific Fama-French Three Factor Model

	Correlations of Unadjusted Momentum Returns							Correlation of Fitted Values							Correlation of Model Residuals							Cov. Decomposition		
	All		Significant Pairs					All		Significant Pairs					All		Significant Pairs					All		
	Avg	t-stat	Avg	t-stat	n	Range		Avg	t-stat	Avg	t-stat	n	Range		Avg	t-stat	Avg	t-stat	n	Range		% Fit	% Err	% Other
Australia	0.18	10.24	0.23	18.52	8	0.18 0.28		0.16	8.22	0.19	17.98	10	0.14 0.24		0.07	2.83	0.22	5.62	3	0.17 0.30		0.32	0.42	0.26
Austria	0.26	10.42	0.33	19.58	8	0.28 0.40		0.09	2.00	0.43	6.71	2	0.37 0.49		0.10	2.64	0.29	26.07	4	0.26 0.31		0.18	0.30	0.52
Belgium	0.18	8.75	0.24	12.58	9	0.17 0.34		0.13	5.36	0.22	14.70	6	0.16 0.26		0.04	1.69	0.18		1	0.18 0.18		0.32	0.28	0.40
Canada																								
Denmark	0.26	8.57	0.32	14.24	13	0.20 0.45		0.29	10.51	0.37	15.91	11	0.20 0.47		0.14	5.10	0.25	15.99	6	0.21 0.30		0.50	0.44	0.06
Finland	0.28	6.63	0.46	19.24	5	0.41 0.55		0.28	7.37	0.50	24.73	4	0.47 0.55		0.14	3.52						0.57	0.43	0.00
France	0.32	12.76	0.34	12.14	15	0.19 0.53		0.30	7.28	0.35	9.77	15	0.14 0.52		0.21	7.33	0.29	15.07	10	0.22 0.43		0.44	0.42	0.15
Germany	0.32	11.17	0.35	13.96	15	0.18 0.51		0.33	8.28	0.36	10.50	16	0.18 0.63		0.16	8.21	0.23	15.24	7	0.18 0.28		0.61	0.37	0.02
Hong Kong	0.18	7.95	0.21	9.33	13	0.12 0.43		0.13	5.59	0.20	8.56	5	0.13 0.26		0.07	3.42	0.24		1	0.24 0.24		0.26	0.46	0.28
Ireland	0.31	11.91	0.35	13.62	13	0.23 0.51		0.10	4.17	0.24		1	0.24 0.24		0.15	4.95	0.31	6.27	4	0.24 0.45		0.22	0.36	0.42
Italy	0.27	9.02	0.32	10.70	13	0.17 0.53		0.29	6.95	0.36	9.46	13	0.18 0.63		0.15	4.50	0.28	8.84	8	0.17 0.45		0.40	0.50	0.10
Japan	0.19	11.13	0.22	15.23	13	0.12 0.29		0.13	4.77	0.23	5.18	7	0.14 0.47		0.11	4.73	0.20	15.10	6	0.17 0.24		0.34	0.30	0.36
Luxembourg																								
Netherlands	0.32	10.84	0.36	13.62	15	0.18 0.55		0.27	8.04	0.32	8.44	12	0.14 0.60		0.23	7.25	0.31	10.46	11	0.18 0.50		0.28	0.65	0.07
New Zealand																								
Norway	0.26	7.92	0.43	13.03	3	0.38 0.49		0.38	7.06	0.59	21.00	8	0.47 0.74		0.11	4.19						0.68	0.34	-0.02
Singapore	0.16	6.07	0.27	7.43	6	0.19 0.43		0.12	2.87	0.27	4.22	6	0.18 0.59		0.09	4.55	0.18	31.67	3	0.17 0.19		0.26	0.37	0.37
Spain	0.17	6.03	0.26	6.59	8	0.17 0.49		0.20	8.42	0.29	16.29	9	0.20 0.35		0.03	1.28						0.61	0.26	0.13
Sweden	0.28	10.10	0.31	11.46	14	0.15 0.44		0.21	6.86	0.30	12.56	9	0.18 0.40		0.18	9.09	0.24	16.53	8	0.18 0.30		0.34	0.53	0.13
Switzerland	0.33	12.06	0.35	12.76	16	0.15 0.51		0.30	6.21	0.38	8.89	13	0.16 0.74		0.17	6.94	0.27	9.92	7	0.18 0.39		0.36	0.39	0.25
UK	0.35	15.53	0.35	14.62	17	0.21 0.51		0.28	8.14	0.34	9.42	13	0.13 0.56		0.19	6.77	0.26	8.77	10	0.18 0.50		0.45	0.40	0.15
USA	0.32	11.80	0.33	11.29	14	0.14 0.48		0.22	6.38	0.29	9.94	12	0.19 0.53		0.12	3.81	0.22	15.62	5	0.18 0.26		0.49	0.36	0.15
Total	0.26	9.94	0.32	13.15				0.22	6.55						0.13	4.88						0.40	0.40	0.20
1975-1980	0.13	2.78	0.40	24.64				0.12	1.35						0.07	1.86						0.23	0.33	0.44
1981-1985	0.07	1.15	0.36	15.64				0.07	1.08						0.06	1.17						0.17	0.71	0.12
1986-1990	0.06	1.09	0.37	15.50				0.05	0.85						0.03	0.51						0.32	0.48	0.20
1991-1995	0.21	7.53	0.33	11.87				0.24	5.98						0.13	4.62						0.31	0.51	0.18
1996-2002	0.33	9.02	0.42	13.60				0.33	7.34						0.18	5.48						0.40	0.43	0.17

Table VI continued

We report average correlations of risk-adjusted momentum returns from 1975-2002 for developed countries. In Panel A, we use a country-specific market model, while in Panel B we use a country-specific model similar to that of Fama-French. For the country-specific market model, we use the equal weighted average of all stocks' local currency return within the country where there are three or more firms. For the Fama-French three factor country-specific model, we use the above market factor along with an SMB factor calculated using NYSE 30/70 breakpoints converted to local currency. Small and Big portfolios is value weighted and must have at least three stocks. The SMB factor is the simple difference between the two portfolios. The HML factor is from Ken French's website. The last three columns decompose the pairwise covariance of country-neutral momentum into the Percent Fit + Percent Error + Percent Other. Specifically, $\text{Cov}(R_A, R_B) = \text{Cov}(R_A^{Fit} + R_A^{Err}, R_B^{Fit} + R_B^{Err}) = \text{Cov}(R_A^{Fit}, R_B^{Fit}) + \text{Cov}(R_A^{Err}, R_B^{Err}) + [\text{Cov}(R_A^{Fit}, R_B^{Err}) + \text{Cov}(R_A^{Err}, R_B^{Fit})]$, where R_A is the country-neutral momentum of country A and R_A^{Fit} is the portion of the return explained by the model, and R_A^{Err} is the portion not explained by the model.

Table VII**Predicting Momentum Portfolio Comovement with Factor Comovement (1975 – 2002)**

Panel A: Regressions of Country-Neutral Momentum Return Covariances on Factor Covariances for Developed Countries

Variables	Full Sample Period				Sub-Periods	
	1975-2002				1975-1988	1989-2002
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	3.23 (3.59)	4.78 (15.20)	4.80 (15.39)	3.75 (4.45)	0.65 (0.78)	5.10 (4.84)
$R_{MKT} - R_f$	0.19 (3.65)			0.04 (0.66)	0.04 (0.71)	0.02 (0.27)
SMB		0.28 (5.95)		0.19 (3.47)	-0.21 (-1.78)	0.20 (3.11)
HML			0.36 (6.35)	0.27 (4.37)	0.15 (0.90)	0.28 (4.17)
Country Pairs:	230	206	189	186	62	186
Adj R^2	0.05	0.14	0.17	0.23	0.03	0.20

Panel B: Panel Regressions of Country-Neutral Momentum Return Covariances during the Holding Period on Covariances in Factor Returns during the Measurement Period for Developed Countries

Variables	Full Sample Period			Sub-Periods	
	1975-2002			1975-1988	1989-2002
	(1)	(2)	(3)	(4)	(5)
Constant	37.66 (11.19)	41.91 (8.45)	3.57 (0.76)	-2.53 (-0.66)	2.71 (0.55)
$R_{MKT} - R_f$	0.16 (12.16)	0.18 (9.91)	0.12 (6.42)	0.05 (2.14)	0.16 (6.05)
SMB		0.22 (8.94)	0.22 (8.61)	0.01 (0.11)	0.24 (8.29)
HML			0.12 (7.03)	0.16 (2.90)	0.11 (5.89)
Country Pairs: Months:	231	187	152 313	45	152
Adj R^2	0.19	0.25	0.24	0.10	0.26

Table VII continued

For Panel A, we calculate the sample covariance of country-neutral momentum returns and country specific risk factors. We then regress the covariance of momentum returns on the covariance of risk factors for all pairs of developed countries that meet the sample selection criteria, have local factors available, and have a valid momentum return on or within 12 months of the factor realization. For Panel B, we examine if the covariance in factor returns during the measurement period predict the covariance of momentum returns in the holding period. For each country-neutral momentum strategy and for each local factor realization, we calculate the following deviations from the country time series mean:

$$\begin{aligned}\varepsilon_{i,t} &= R_{i,t} - \bar{R}_i \\ \gamma_{i,t}^{RMRF} &= RMRF_{i,t} - \overline{RMRF}_i\end{aligned}$$

where $R_{i,t}$ is the country-neutral momentum return in country i in month t , and $RMRF_{i,t}$ is the return to the excess local market factor in country i in month t . We do similar calculations for SMB and HML. We then calculate the cross product for country i and j :

$$\begin{aligned}y_{i,j,t} &= \varepsilon_{i,t} \cdot \varepsilon_{j,t} \\ x_{i,j,t} &= \left(\frac{1}{11}\right) \cdot \sum_{k=2}^{12} \left(\gamma_{i,t-k}^{RMRF} \cdot \gamma_{j,t-k}^{RMRF} \right)\end{aligned}$$

For the regression analysis, we use panel regressions of y on x and include time and country-pair dummies.